# Solutions to the Exercises of Chapter 6

(Phase Plane)

Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed.

**Conventions and notes.** Dots denote $d/dt$. Problem statements are paraphrased briefly; refer to the book for the exact wording. For a fixed point of $\dot{x} = f, \dot{y} = g$, $A$ is the Jacobian, $\tau = \operatorname{tr} A$, $\Delta = \det A$; the eigenvalues solve $\lambda^2 - \tau\lambda + \Delta = 0$. In the phase portraits (generated numerically) streamlines are blue, the $\dot{x} = 0$ nullclines are dashed red, the $\dot{y} = 0$ nullclines are dotted green, and black dots are fixed points. Problems that ask for a "computer-generated" portrait are answered with such a figure. A few points where the printed exercise needed interpretation (6.4.9, 6.6.11 in particular) are flagged where they occur.

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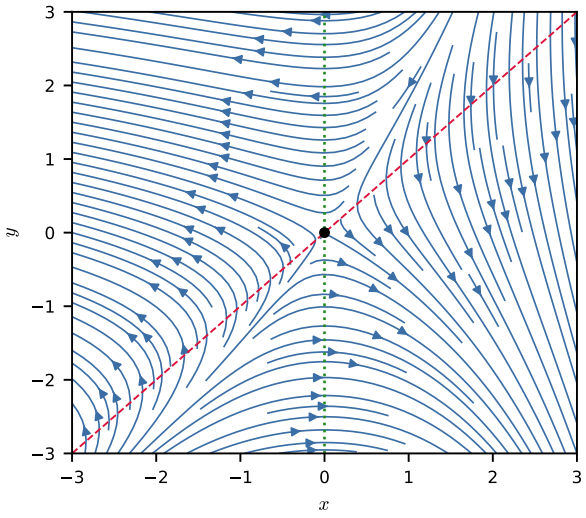
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## 6.1 Phase Portraits

### 6.1.1 $\dot{x} = x - y,\ \dot{y} = 1 - e^x$

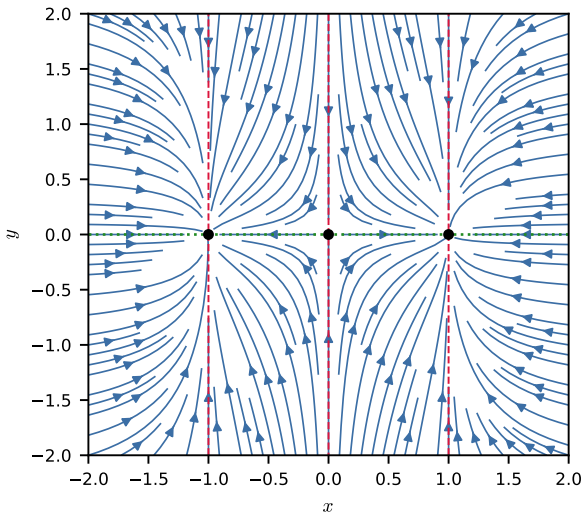
**Solution.** Fixed points: $\dot{y} = 0 \Rightarrow x = 0$, then $\dot{x} = 0 \Rightarrow y = 0$. So $(0, 0)$ only. The nullclines are the line $y = x$ ($\dot{x} = 0$) and the line $x = 0$ ($\dot{y} = 0$). On $x = 0$, $\dot{x} = -y$ (flow points left for $y > 0$, right for $y < 0$); on $y = x$, $\dot{y} = 1 - e^x$ (up for $x < 0$, down for $x > 0$). Jacobian $A = \begin{pmatrix} 1 & -1 \\ -e^x & 0 \end{pmatrix} \to \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix}$, $\Delta = -1 < 0$: a *saddle*, $\lambda = (1 \pm \sqrt{5})/2$. Unstable direction $(1, -0.62)$, stable direction $(1, 1.62)$.




6.1.1: single saddle.

### 6.1.2 $\dot{x} = x - x^3,\ \dot{y} = -y$

**Solution.** Fixed points: $y = 0$, $x \in \{0, \pm 1\}$. $A = \mathrm{diag}(1 - 3x^2, -1)$. At $(0, 0)$: $\mathrm{diag}(1, -1)$, saddle. At $(\pm 1, 0)$: $\mathrm{diag}(-2, -1)$, stable nodes. The system is decoupled: $y \to 0$, and $x \to \pm 1$ according to the sign of $x_0$; the $y$-axis (stable manifold of the origin) separates the two basins. Nullclines: vertical lines $x = 0, \pm 1$ and the $x$-axis.



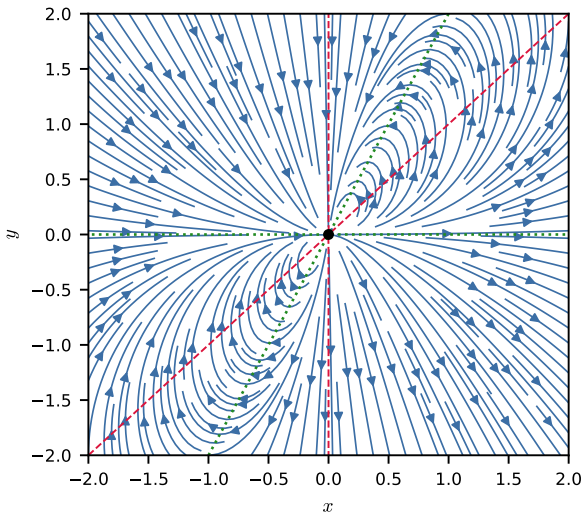

6.1.2.

### 6.1.3 $\dot{x} = x(x-y),\ \dot{y} = y(2x-y)$

**Solution.** Fixed points: if $x = 0$ then $y^2 = 0$; if $y = x$ then $\dot{y} = x^2 = 0$. Only the origin. The Jacobian there is the zero matrix, so linearization says nothing. Use polar coordinates:

$$\dot{r} = r^2\big(\cos^3\theta - \cos^2\theta\sin\theta + 2\sin^2\theta\cos\theta - \sin^3\theta\big), \qquad \dot{\theta} = r\sin\theta\cos^2\theta.$$

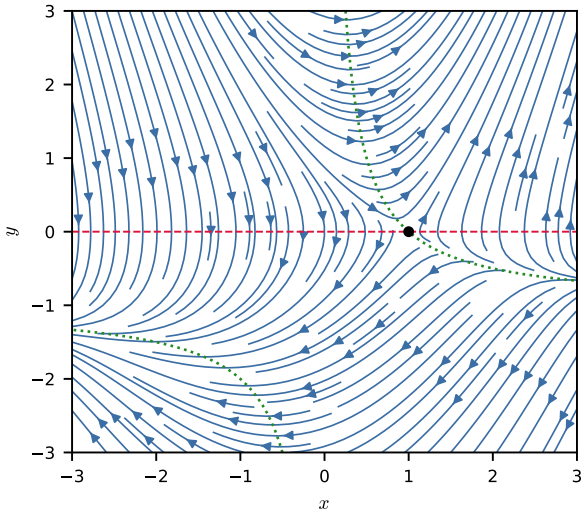
The rays $\theta = 0, \pi$ ($x$-axis) and $\theta = \pm\pi/2$ ($y$-axis) are invariant. On them: $\dot{r} = r^2$ at $\theta = 0$ (outward), $-r^2$ at $\theta = \pi$ and $\theta = \pi/2$ (inward), $+r^2$ at $\theta = -\pi/2$ (outward). Since $\dot{\theta} \propto \sin\theta$, angles in the upper half plane increase toward $\pi$ (except they stall at $\pi/2$) and in the lower half plane decrease toward $-\pi$. Consequently the origin is a degenerate (non-hyperbolic) point: in the first quadrant trajectories leave the origin tangent to the $+x$ axis and return to it tangent to the $+y$ axis (loops); in the second quadrant they come in from infinity near the $+y$ axis and enter the origin tangent to the $-x$ axis; in the third quadrant they leave tangent to the $-y$ axis and enter tangent to the $-x$ axis; in the fourth quadrant they leave tangent to the $+x$ axis and escape to infinity along the $-y$ axis. (Because the field is homogeneous quadratic, the picture is self-similar under $(x, y) \mapsto \lambda(x, y)$.)




6.1.3.

### 6.1.4 $\dot{x} = y,\ \dot{y} = x(1+y) - 1$

**Solution.** Fixed point: $y = 0$, $x = 1$. $A = \begin{pmatrix} 0 & 1 \\ 1+y & x \end{pmatrix} \to \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $\Delta = -1$: saddle. Nullclines: $y = 0$ and the hyperbola $x = 1/(1+y)$.

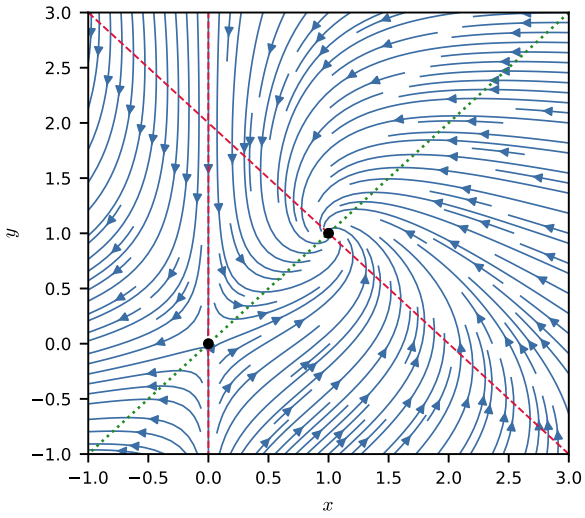



6.1.4.

### 6.1.5 $\dot{x} = x(2 - x - y),\ \dot{y} = x - y$

**Solution.** $x = 0 \Rightarrow y = 0$; $x \neq 0 \Rightarrow y = 2 - x$ and $y = x$, so $(1, 1)$. Fixed points $(0, 0), (1, 1)$. $A = \begin{pmatrix} 2 - 2x - y & -x \\ 1 & -1 \end{pmatrix}$. At $(0, 0)$: $\begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$, eigenvalues $2, -1$: saddle (stable manifold = $y$-axis,

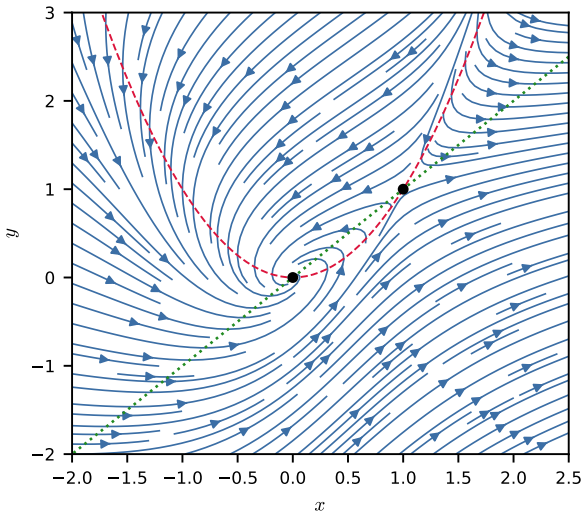
unstable direction $(3,1)$). At $(1,1)$: $\begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}$, $\tau = -2, \Delta = 2$, $\lambda = -1 \pm i$: stable spiral.




6.1.5.

### 6.1.6 $\dot{x} = x^2 - y,\ \dot{y} = x - y$

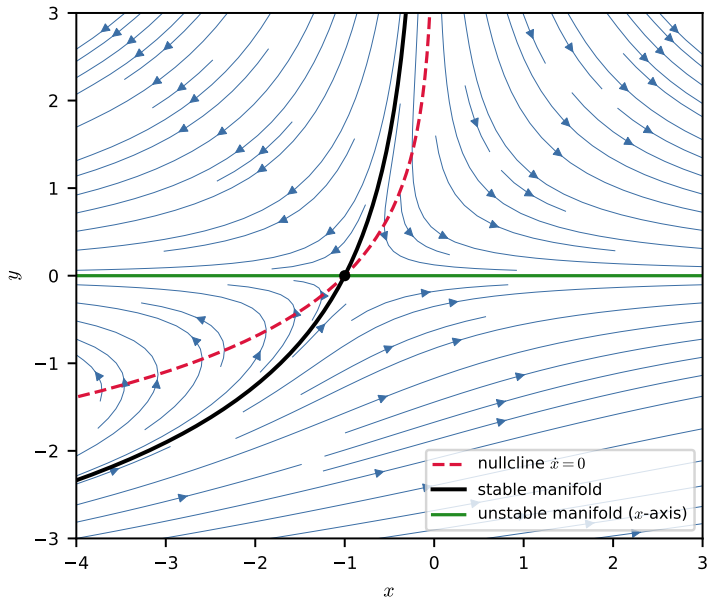
**Solution.** $y = x^2$ and $y = x$ give $(0,0)$ and $(1,1)$. $A = \begin{pmatrix} 2x & -1 \\ 1 & -1 \end{pmatrix}$. At $(0,0)$: $\tau = -1, \Delta = 1$, $\tau^2 - 4\Delta < 0$: stable spiral. At $(1,1)$: $\tau = 1, \Delta = -1$: saddle.




6.1.6.

### 6.1.7 *Nullcline vs. stable manifold in Example 6.1.1:* $\dot{x} = x + e^{-y},\ \dot{y} = -y$

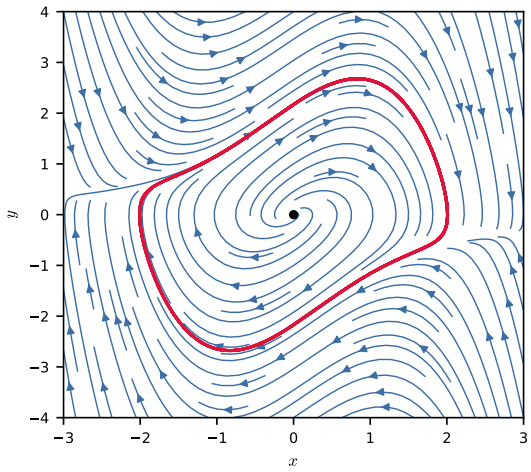
**Solution.** The nullcline $\dot{x} = 0$ is $x = -e^{-y}$. Near the saddle $(-1, 0)$, with $u = x + 1$: $u = 1 - e^{-y} \approx y - \frac{1}{2}y^2$, so the nullcline has slope $dy/du = 1$ at the saddle. The stable manifold is tangent to the stable eigenvector of $\begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}$ for $\lambda = -1$, which is $(1, 2)$: slope 2 (exercise 6.1.14 gives $y = 2u + \frac{4}{3}u^2 + \dots$). So the curves differ. The conceptual difference: on the nullcline the flow is *vertical* (crossing it), whereas the stable manifold is an *invariant* curve, tangent to the flow. Where they sit relative to each other: on the stable manifold, $\dot{x} = -u < 0$ for $u > 0$, i.e. the manifold is to the *left* of the nullcline in the upper half-plane (and by the same token below the nullcline in the lower half plane).




6.1.7: both curves through the saddle (black = stable manifold, dashed red = nullcline).

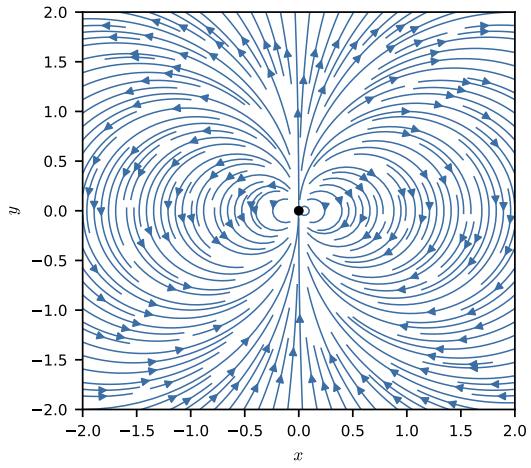
### 6.1.8–6.1.11 *Computer phase portraits*

**Solution.** *6.1.8 (van der Pol, $\mu = 1$).* Only fixed point: origin with $A = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$, $\tau = 1, \Delta = 1$: unstable spiral. All trajectories (except the origin) approach a unique stable limit cycle (thick red curve, amplitude $\approx 2$ in $x$).



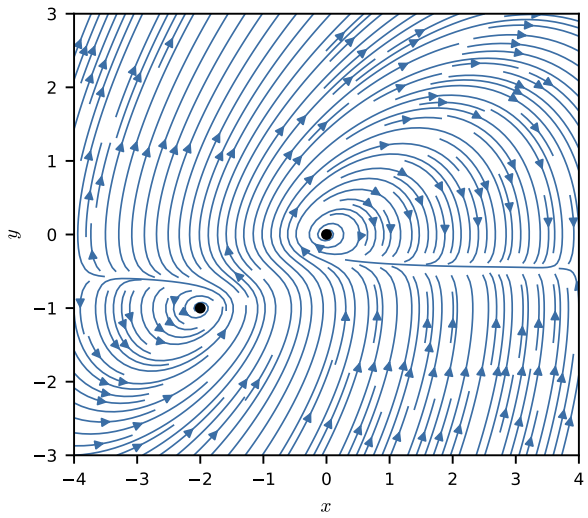

6.1.8: van der Pol oscillator.

*6.1.9 (dipole).* Only the origin is fixed. Writing $v = y^2$: $dy/dx = (y^2 - x^2)/(2xy)$ gives $d(y^2)/dx - y^2/x = -x$, so $(y^2/x)' = -1$ and $x^2 + y^2 = Cx$: all trajectories are circles through the origin tangent to the $y$-axis (clockwise for $x > 0$, counter-clockwise for $x < 0$), and the $y$-axis itself is the flow line $\dot{y} = y^2$. Every orbit is homoclinic to the origin.



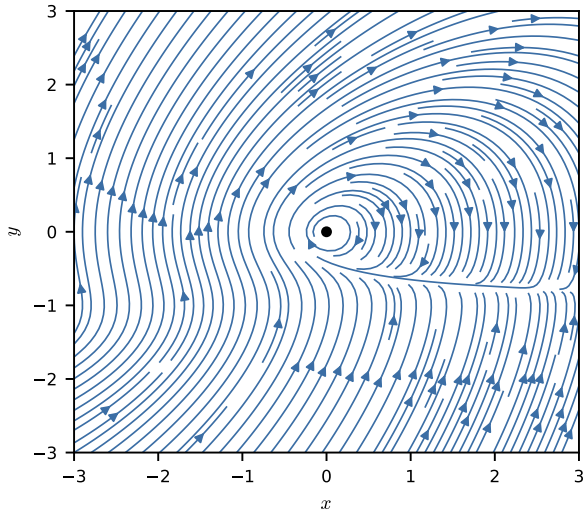

6.1.9.

*6.1.10 (two-eyed monster).* $\dot{x} = 0 \Rightarrow y = 0$ or $-1$. $y = 0 \Rightarrow x = 0$; $y = -1 \Rightarrow x/2+1 = 0 \Rightarrow x = -2$. Fixed points $(0, 0)$ with $A = \begin{pmatrix} 0 & 1 \\ -1/2 & 1/5 \end{pmatrix}$ ($\tau = \frac{1}{5}, \Delta = \frac{1}{2}$: unstable spiral) and $(-2, -1)$ with $A = \begin{pmatrix} 0 & -1 \\ 1/2 & -1/5 \end{pmatrix}$ ($\tau = -\frac{1}{5}, \Delta = \frac{1}{2}$: stable spiral): the two "eyes".




6.1.10.

*6.1.11 (parrot).* $y = 0 \Rightarrow x = 0$; $y = -1$ gives $-x - \frac{1}{5} + x + \frac{6}{5} = 1 \neq 0$. Only fixed point is the origin, an unstable spiral ($\tau = \frac{1}{5}, \Delta = 1$); most trajectories escape to infinity.

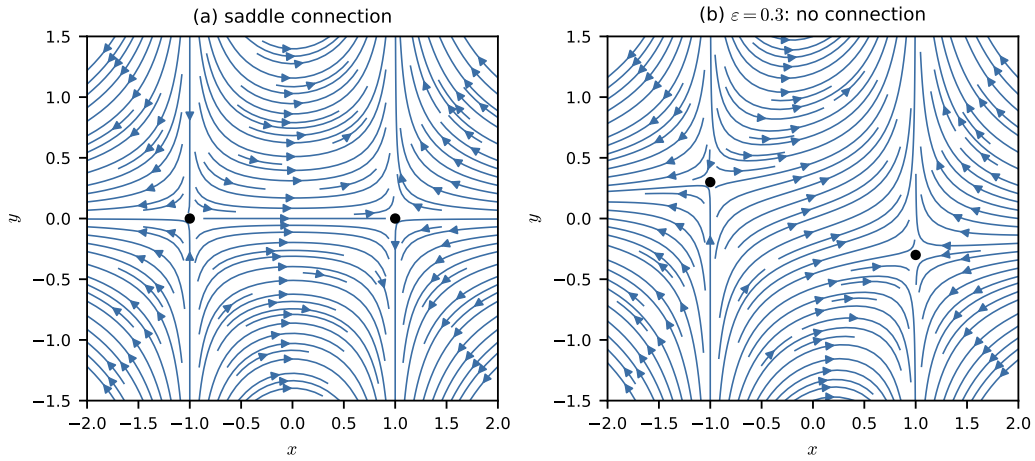



6.1.11.

### 6.1.12 *Saddle connections*

**Solution.** (a) Take $\dot{x} = 1 - x^2$, $\dot{y} = xy$. Fixed points $(\pm 1, 0)$ with $A = \begin{pmatrix} -2x & 0 \\ y & x \end{pmatrix}$: at $(1,0)$ eigenvalues $-2, 1$; at $(-1,0)$ eigenvalues $2, -1$: both saddles. The $x$-axis is invariant and $\dot{x} = 1 - x^2 > 0$ on $|x| < 1$, so the segment of the $x$-axis is a trajectory from $(-1,0)$ to $(1,0)$. (b) Perturb to $\dot{y} = xy + \varepsilon$. Now the saddles sit at $(1, -\varepsilon)$ and $(-1, \varepsilon)$. Since $\dot{x}$ does not involve $y$, $dy/dx = (xy + \varepsilon)/(1 - x^2)$, whose solution leaving $(-1, \varepsilon)$ is $y = \varepsilon\,(\arcsin x + \pi/2)/\sqrt{1 - x^2} \to \pm\infty$

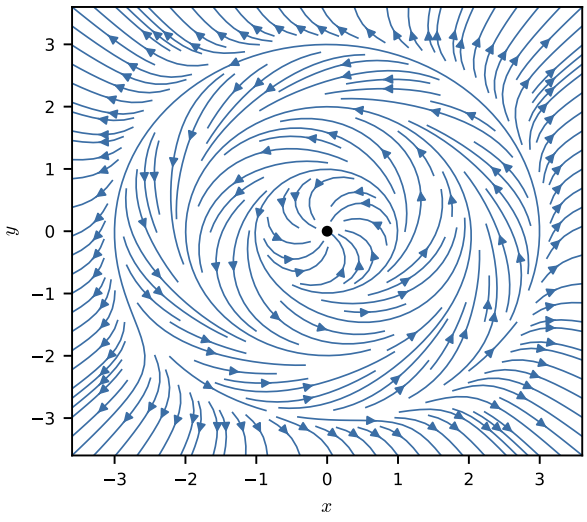
as $x \to 1$. So the unstable manifold of $(-1, \varepsilon)$ misses $(1, -\varepsilon)$: no connection.




6.1.12: (a) connection, (b) no connection.

### 6.1.13 *Exactly three closed orbits and one fixed point*

**Solution.** Nested circles around a fixed point. For instance $\dot{\theta} = 1$, $\dot{r} = r(r-1)(r-2)(r-3)$: closed orbits $r = 1, 2, 3$, one fixed point (stable focus at the origin). The circles $r = 1, 3$ are unstable, $r = 2$ stable. (A closed orbit must enclose a fixed point, so all three must be nested around the single fixed point; Section 6.8.)




6.1.13.

### 6.1.14 *Series for the stable manifold of the saddle in $\dot{x} = x + e^{-y}$, $\dot{y} = -y$*

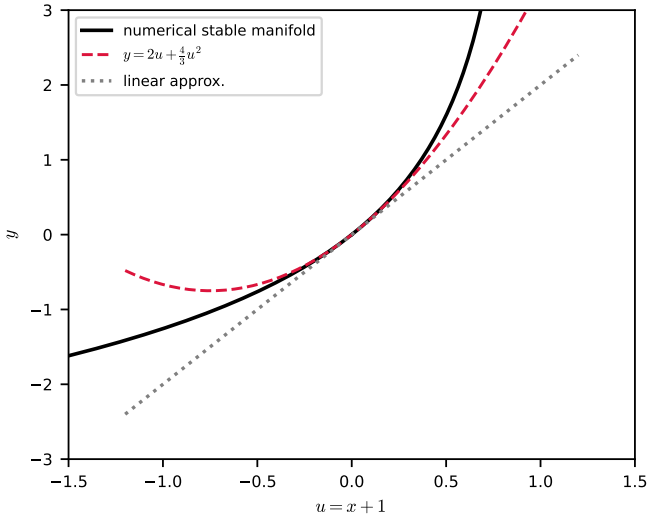
**Solution.** (a) Let $u = x + 1$. Then $\dot{u} = u - 1 + e^{-y} = u - y + \frac{1}{2}y^2 - \dots$, $\dot{y} = -y$. Put $y = a_1 u + a_2 u^2 + O(u^3)$. First expression: $dy/du = a_1 + 2a_2 u + \dots$. Second:

$$\frac{dy}{du} = \frac{\dot{y}}{\dot{u}} = \frac{-y}{u - y + \frac{1}{2}y^2 + \dots} = \frac{-(a_1 + a_2 u)}{(1 - a_1) + u\big(\frac{1}{2}a_1^2 - a_2\big) + \dots}.$$

Order $u^0$: $a_1(1 - a_1) = -a_1 \Rightarrow a_1 = 0$ or $a_1 = 2$; the stable manifold has $a_1 = 2$ (slope of the stable eigenvector). Order $u^1$ with $a_1 = 2$: denominator $= -1 + u(2 - a_2)$, so $\frac{dy}{du} = (2 + a_2 u)(1 + (2 - a_2)u) + \dots = 2 + (4 - a_2)u$. Equating with $2 + 2a_2 u$: $2a_2 = 4 - a_2$, $a_2 = \frac{4}{3}$. Hence

$$\boxed{y = 2u + \tfrac{4}{3}u^2 + O(u^3), \qquad u = x + 1.}$$

(b) The curve rises steeply to the upper right (convex) and goes down to the lower left, flattening as it goes, matching the shape in Figure 6.1.4; the figure compares it with the numerically computed manifold (integrating backward from the saddle).




6.1.14: series (dashed red) vs. numerical stable manifold (black).

## 6.2 Existence, Uniqueness, and Topological Consequences

### 6.2.1 *Trajectories appearing to intersect at a fixed point*

**Solution.** No contradiction. A fixed point is itself a trajectory (a constant solution). Uniqueness says two *different* trajectories cannot share a point; a trajectory that approaches a fixed point does so only as $t \to \pm\infty$ (in finite time it cannot reach it, by uniqueness applied to the constant solution), so it never actually contains the fixed point.

### 6.2.2 $\dot{x} = y,\ \dot{y} = -x + (1 - x^2 - y^2)y$

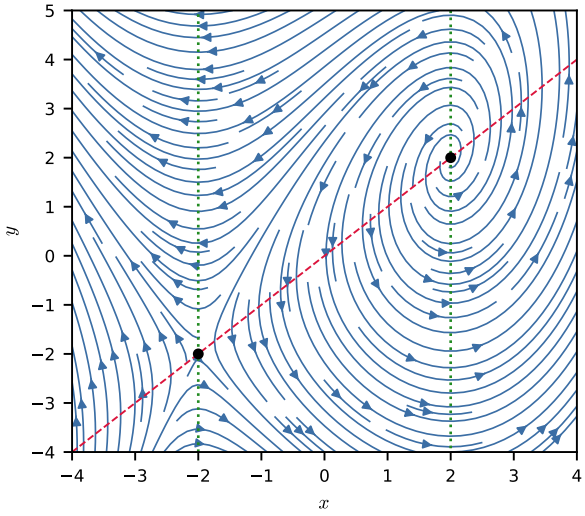
**Solution.** (a) The vector field is polynomial, hence $C^1$ on all of $\mathbb{R}^2$ (in particular on $D = \{x^2 + y^2 < 4\}$); the existence–uniqueness theorem applies. (b) With $x = \sin t$, $y = \cos t$: $\dot{x} = \cos t = y$ ✓;

$\dot{y} = -\sin t = -x$ and $(1 - \sin^2 t - \cos^2 t)y = 0$, so $\dot{y} = -x + 0$ ✓. (c) The unit circle is a closed trajectory. Trajectories cannot cross it (uniqueness), so a trajectory starting inside ($r = 1/2 < 1$) stays inside for all time, forward and backward: $x(t)^2 + y(t)^2 < 1$ for all $t$.

## 6.3 Fixed Points and Linearization

### 6.3.1 $\dot{x} = x - y,\ \dot{y} = x^2 - 4$

**Solution.** $x = \pm 2$, $y = x$: fixed points $(2, 2), (-2, -2)$. $A = \begin{pmatrix} 1 & -1 \\ 2x & 0 \end{pmatrix}$. At $(2, 2)$: $\tau = 1, \Delta = 4$, $\tau^2 - 4\Delta < 0$: unstable spiral. At $(-2, -2)$: $\Delta = -4$: saddle, $\lambda = (1 \pm \sqrt{17})/2$.



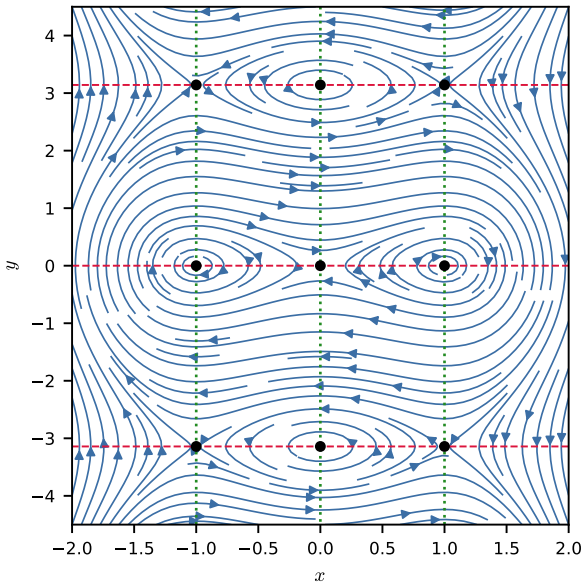

6.3.1 (also the computer check of 6.3.7).

### 6.3.2 $\dot{x} = \sin y,\ \dot{y} = x - x^3$

**Solution.** Fixed points: $y = n\pi$, $x \in \{0, \pm 1\}$. $A = \begin{pmatrix} 0 & \cos y \\ 1 - 3x^2 & 0 \end{pmatrix}$, $\tau = 0$, $\Delta = -(-1)^n(1 - 3x^2)$.

- $x = 0$: $\Delta = -(-1)^n$: saddle for $n$ even, center for $n$ odd.
- $x = \pm 1$: $\Delta = 2(-1)^n$: center for $n$ even, saddle for $n$ odd.

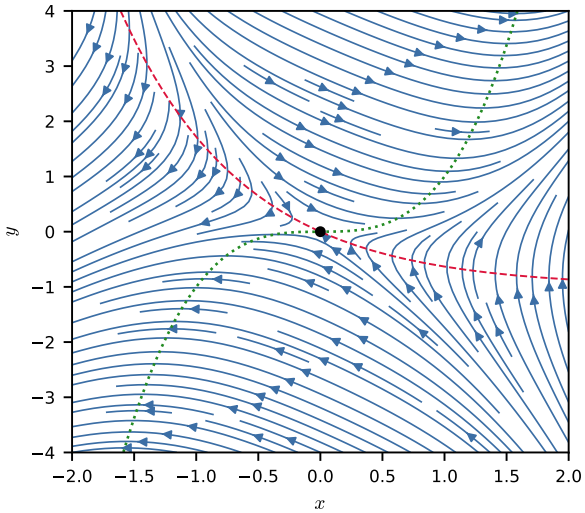
The system is conservative: $H = -\cos y - \frac{1}{2}x^2 + \frac{1}{4}x^4$ satisfies $\dot{H} = 0$, and the "centers" are strict local extrema of $H$, hence true nonlinear centers.




6.3.2.

### 6.3.3 $\dot{x} = 1 + y - e^{-x},\ \dot{y} = x^3 - y$

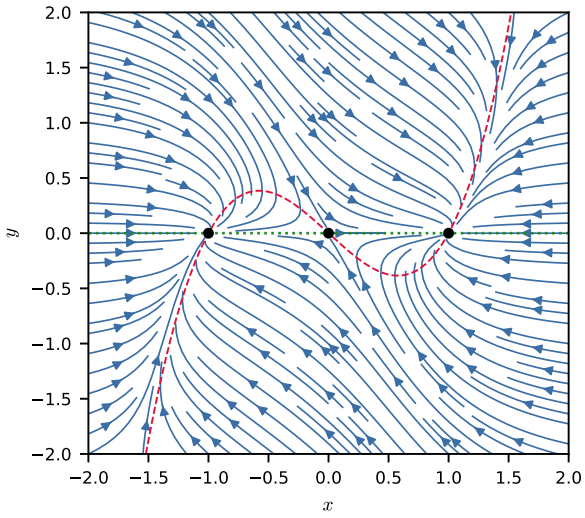
**Solution.** $y = x^3$, so $g(x) = 1 + x^3 - e^{-x} = 0$. Since $g'(x) = 3x^2 + e^{-x} > 0$, $g$ is strictly increasing and its only root is $x = 0$. Single fixed point $(0, 0)$; $A = \begin{pmatrix} e^{-x} & 1 \\ 3x^2 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$: eigenvalues $\pm 1$, a saddle (unstable direction $(1, 0)$, stable direction $(1, -2)$).




6.3.3.

### 6.3.4 $\dot{x} = y + x - x^3,\ \dot{y} = -y$

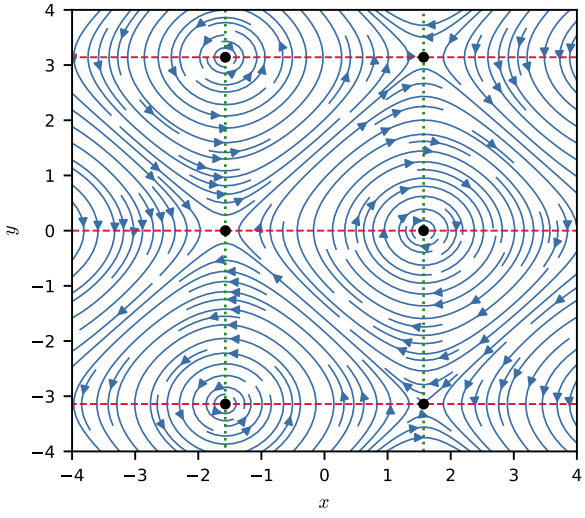
**Solution.** $y = 0$, $x \in \{0, \pm 1\}$. $A = \begin{pmatrix} 1 - 3x^2 & 1 \\ 0 & -1 \end{pmatrix}$. At the origin: eigenvalues $1, -1$: saddle (unstable direction $(1, 0)$, stable direction $(1, -2)$). At $(\pm 1, 0)$: eigenvalues $-2, -1$: stable nodes; trajectories enter tangent to the slow direction $(1, 1)$ ($\lambda = -1$).




6.3.4.

### 6.3.5 $\dot{x} = \sin y,\ \dot{y} = \cos x$

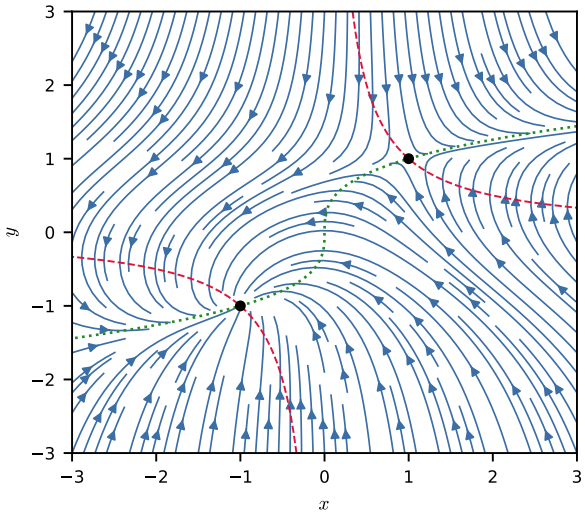
**Solution.** Fixed points $(\frac{\pi}{2} + m\pi,\ n\pi)$. $A = \begin{pmatrix} 0 & \cos y \\ -\sin x & 0 \end{pmatrix}$, $\tau = 0$, $\Delta = \sin x \cos y = (-1)^{m+n}$. So $m+n$ even: centers; $m+n$ odd: saddles (checkerboard). Conservative with $H = -\cos y - \sin x$, so the centers are nonlinear centers.




6.3.5.

### 6.3.6 $\dot{x} = xy - 1,\ \dot{y} = x - y^3$

**Solution.** $x = 1/y$ and $x = y^3 \Rightarrow y^4 = 1$: fixed points $(1,1), (-1,-1)$. $A = \begin{pmatrix} y & x \\ 1 & -3y^2 \end{pmatrix}$. At $(1,1)$: $\begin{pmatrix} 1 & 1 \\ 1 & -3 \end{pmatrix}$, $\Delta = -4$: saddle ($\lambda = -1 \pm \sqrt{5}$). At $(-1,-1)$: $\begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix}$, $\tau = -4, \Delta = 4, \tau^2 - 4\Delta = 0$: repeated eigenvalue $-2$ with a single eigenvector $(1,1)$: a stable degenerate node. (A borderline case: nonlinear terms could turn it into a stable spiral, but it is certainly stable.)




6.3.6.

### 6.3.7 *Computer plots for 6.3.1–6.3.6*

**Solution.** The figures above are the computer-generated portraits (streamlines of the full nonlinear field). They agree with the sketches implied by the linearizations: saddles with the predicted eigendirections, the spiral/node classifications, and the checkerboard of centers and saddles in 6.3.2 and 6.3.5.

### 6.3.8 *Gravitational equilibrium*

**Solution.** (a) Put $m_1$ at 0 and $m_2$ at $a$; the particle is at $x \in (0, a)$. Per unit mass, $m_1$ pulls toward 0: $-Gm_1/x^2$; $m_2$ pulls toward $a$: $+Gm_2/(a-x)^2$. So $\ddot{x} = \dfrac{Gm_2}{(x-a)^2} - \dfrac{Gm_1}{x^2}$. (b) Equilibrium: $m_2 x^2 = m_1 (a-x)^2 \Rightarrow x/(a-x) = \sqrt{m_1/m_2}$, i.e.

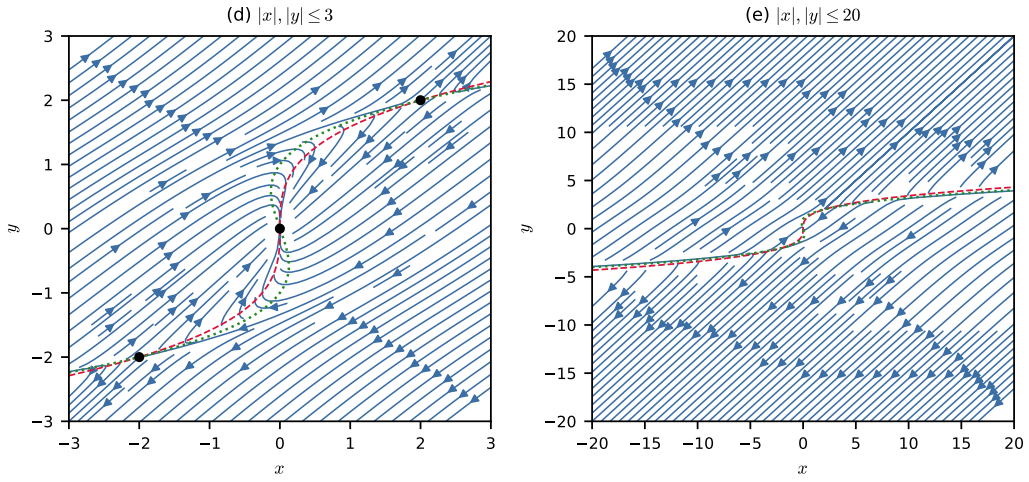
$$x^* = \frac{a\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}}.$$

Writing $\ddot{x} = -V'(x)$ with $V = -Gm_1/x - Gm_2/(a-x)$, $V'' = -2Gm_1/x^3 - 2Gm_2/(a-x)^3 < 0$: $x^*$ is a maximum of the potential, so the equilibrium is *unstable* (a saddle in the $(x, \dot{x})$ plane, eigenvalues $\pm\sqrt{-V''}$).

### 6.3.9 $\dot{x} = y^3 - 4x,\ \dot{y} = y^3 - y - 3x$

**Solution.** (a) Subtracting the equations at a fixed point: $\dot{x} - \dot{y} = y - x = 0$, so $x = y$ and $y^3 = 4y$: $(0,0), (2,2), (-2,-2)$. $A = \begin{pmatrix} -4 & 3y^2 \\ -3 & 3y^2 - 1 \end{pmatrix}$. At $(0,0)$: eigenvalues $-4, -1$: stable node. At $(\pm 2, \pm 2)$: $\begin{pmatrix} -4 & 12 \\ -3 & 11 \end{pmatrix}$, $\Delta = -8$: saddles (eigenvalues $8, -1$). (b) On $x = y$: $\dot{x} = y^3 - 4y$ and $\dot{y} = y^3 - y - 3y = y^3 - 4y$, equal, so $x - y$ stays 0: the line is invariant. (c) $w = x - y$: $\dot{w} = (y^3 - 4x) - (y^3 - y - 3x) = -(x - y) = -w$, so $w = w_0 e^{-t} \to 0$. (d) All trajectories collapse onto the invariant line $x = y$ at rate $e^{-t}$, then move along it according to $\dot{x} = x^3 - 4x$ (stable at 0, unstable at $\pm 2$, escape beyond). (e) The computer picture on $[-20, 20]^2$ shows trajectories hugging an S-shaped curve. Explanation: backward in time ($t \to -\infty$) $w$ grows like $e^{|t|}$ while in the variables $(w, y)$, $\dot{y} = y^3 - 4y - 3w$ has a huge cubic restoring term in reverse time, so $y$ relaxes quickly to the curve $y^3 - 4y - 3w = 0$, i.e. (in terms of $x$) to the nullcline $\dot{y} = 0$:

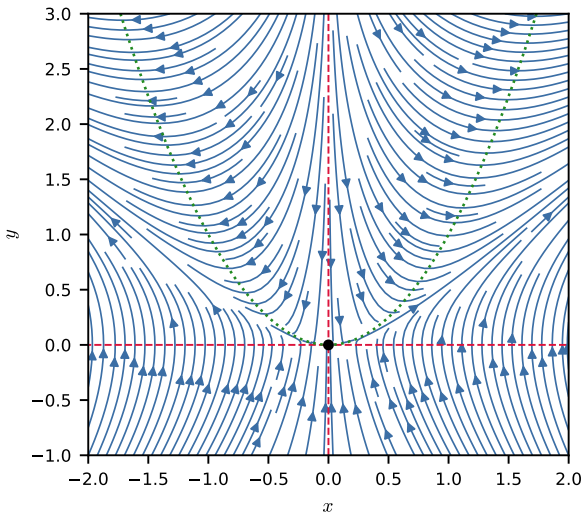
$$x = \tfrac{1}{3}(y^3 - y) \;\approx\; \tfrac{1}{3}y^3 \quad (|y| \gg 1).$$




6.3.9(d),(e).

### 6.3.10 $\dot{x} = xy,\ \dot{y} = x^2 - y$

**Solution.** (a) $A = \begin{pmatrix} y & x \\ 2x & -1 \end{pmatrix} \to \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$: eigenvalue 0 with eigenvector the $x$-direction; the linearized system $\dot{x} = 0, \dot{y} = -y$ has the entire $x$-axis as fixed points (non-isolated). (b) Actual fixed points: $xy = 0$ and $y = x^2$. If $x = 0$ then $y = 0$; if $y = 0$ then $x = 0$. Only the origin: isolated. (c) On the $y$-axis: $\dot{x} = 0, \dot{y} = -y$: invariant, flowing into the origin (stable manifold). Nullcline $y = x^2$: $\dot{y} = 0$, $\dot{x} = x^3$ (right for $x > 0$, left for $x < 0$). On the $x$-axis $\dot{y} = x^2 > 0$. For $y \gg$ anything, $\dot{y} = -(y - x^2)$ attracts to the parabola quickly (a slow manifold $y \approx x^2 - 2x^4 + \dots$), on which $\dot{x} \approx x^3$: motion *away* from the origin on both sides. So the origin is a nonlinear saddle: stable manifold = $y$-axis, unstable (center) manifold = the parabola $y \approx x^2$.




6.3.10.

**6.3.11** $\dot{r} = -r,\ \dot{\theta} = 1/\ln r$

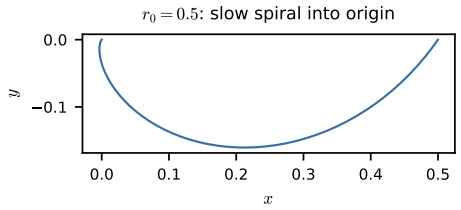
**Solution.** (a) $r(t) = r_0 e^{-t}$ $(0 < r_0 < 1)$. Then $\ln r = \ln r_0 - t$ and $\dot{\theta} = 1/(\ln r_0 - t)$, so

$$\theta(t) = \theta_0 + \ln|\ln r_0| - \ln|\ln r_0 - t|.$$

(b) $r \to 0$ and $|\theta(t)| \approx \ln(t + |\ln r_0|) \to \infty$ (slowly): the orbit winds infinitely often while shrinking to the origin: a stable spiral. (c) $x = r\cos\theta, y = r\sin\theta$: $\dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta = -x - y/\ln r$, $\dot{y} = -y + x/\ln r$, with $\ln r = \frac{1}{2}\ln(x^2 + y^2)$:

$$\dot{x} = -x - \frac{2y}{\ln(x^2+y^2)}, \qquad \dot{y} = -y + \frac{2x}{\ln(x^2+y^2)}.$$

(d) The extra terms are bounded by $r/|\ln r| = o(r)$ as $r \to 0$, so the linearization at the origin is $\dot{x} = -x, \dot{y} = -y$: a stable star. The nonlinear terms, though smaller than linear ones, are enough to turn the star into a spiral.




6.3.11: $r_0 = 0.5$.

### 6.3.12 $\dot\theta = (x\dot y - y\dot x)/r^2$

**Solution.** $\theta = \tan^{-1}(y/x) \Rightarrow \dot\theta = \dfrac{1}{1+y^2/x^2}\cdot\dfrac{x\dot y - y\dot x}{x^2} = \dfrac{x\dot y - y\dot x}{x^2+y^2}$.

### 6.3.13 $\dot x = -y - x^3,\ \dot y = x$

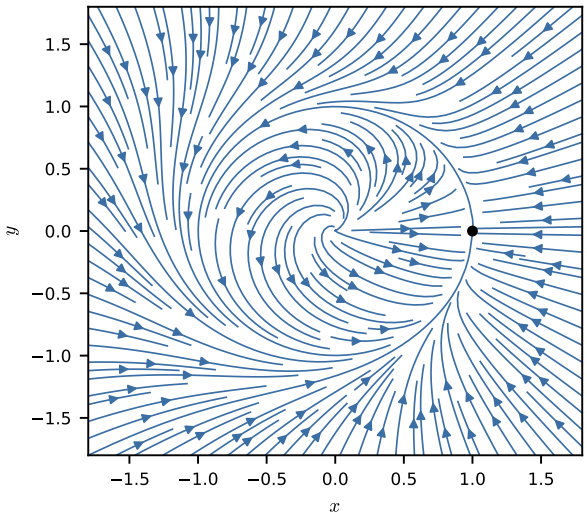
**Solution.** $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$: linear center. In polar form: $r\dot r = x\dot x + y\dot y = -x^4$, so $\dot r = -r^3\cos^4\theta \le 0$; and $\dot\theta = (x\dot y - y\dot x)/r^2 = 1 + r^2\cos^3\theta\sin\theta > 0$ near the origin. So trajectories rotate counterclockwise and $r$ decreases (strictly except at the instants $\cos\theta = 0$). Over one revolution $\Delta r \approx -r^3\int_0^{2\pi}\cos^4\theta\,d\theta = -\frac{3\pi}{4}r^3 < 0$. Equivalently $V = x^2+y^2$ has $\dot V = -2x^4 \le 0$ with equality only on $x = 0$, where the flow is not tangent to the $y$-axis. The origin is a stable spiral.

### 6.3.14 $\dot x = -y + ax^3,\ \dot y = x + ay^3$

**Solution.** Linearization is always a center. Polar: $r\dot r = x\dot x + y\dot y = a(x^4+y^4)$, so $\dot r = ar^3(\cos^4\theta + \sin^4\theta)$ with $\frac{1}{2} \le \cos^4\theta + \sin^4\theta \le 1$; and $\dot\theta = 1 + a\,xy(y^2-x^2)/r^2 \approx 1$. Hence the origin is a *stable spiral for* $a < 0$, an *unstable spiral for* $a > 0$, and (for $a = 0$, a linear system) a *center*.

### 6.3.15 $\dot r = r(1-r^2),\ \dot\theta = 1 - \cos\theta$

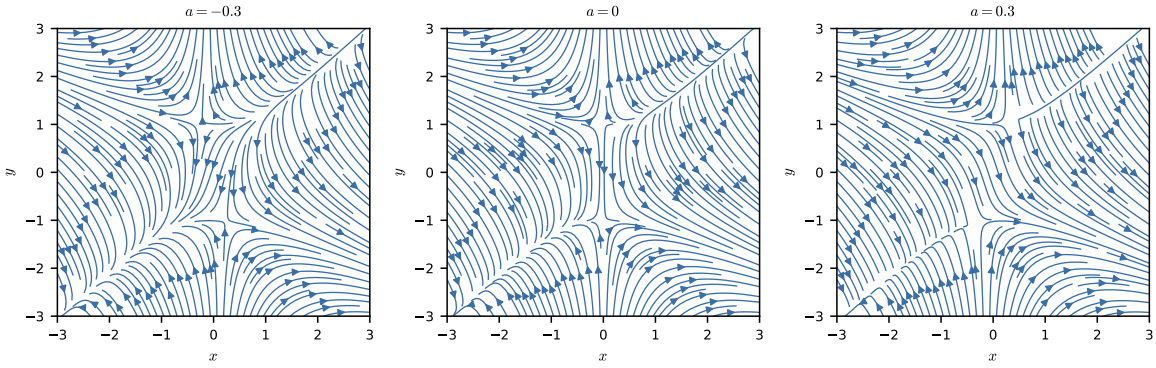
**Solution.** The circle $r = 1$ attracts all $r > 0$ (radial dynamics), and the origin is a repeller. On the circle, $\dot\theta = 1 - \cos\theta \ge 0$ with equality only at $\theta = 0$: the point $(r^*, \theta^*) = (1, 0)$ is the only fixed point there, and the angle always increases around the circle: a point starting at $\theta = \varepsilon > 0$ moves counterclockwise *all the way around* the circle before returning to $\theta = 0$ from below. Thus every trajectory with $r > 0$ tends to $(1, 0)$ (attracting), but trajectories starting arbitrarily close to it make a full excursion of size $\approx 2$ first: not Liapunov stable.




6.3.15.

**6.3.16** $\dot{x} = a + x^2 - xy,\ \dot{y} = y^2 - x^2 - 1$

**Solution.** (a) $a = 0$: $\dot{x} = x(x - y)$. Fixed points: $x = 0 \Rightarrow y = \pm 1$; $y = x \Rightarrow \dot{y} = -1 \neq 0$. So $(0, \pm 1)$; $A = \begin{pmatrix} 2x - y & -x \\ -2x & 2y \end{pmatrix}$ gives $\mathrm{diag}(-1, 2)$ at $(0, 1)$ and $\mathrm{diag}(1, -2)$ at $(0, -1)$: two saddles. The $y$-axis is invariant ($\dot{x} = 0$ there) with $\dot{y} = y^2 - 1 < 0$ for $|y| < 1$: the segment from $(0, 1)$ to $(0, -1)$ is a trajectory: a saddle connection. (b) For $a \neq 0$ the saddles move to $\approx (a, 1)$ and $\approx (-a, -1)$ and the invariance of $x = 0$ is lost: near the $y$-axis $\dot{x} \approx a$, so the unstable manifold of the top saddle drifts to the right of the lower saddle if $a > 0$ (to the left if $a < 0$) and misses it. The connection is destroyed by an arbitrarily small perturbation: not structurally stable.



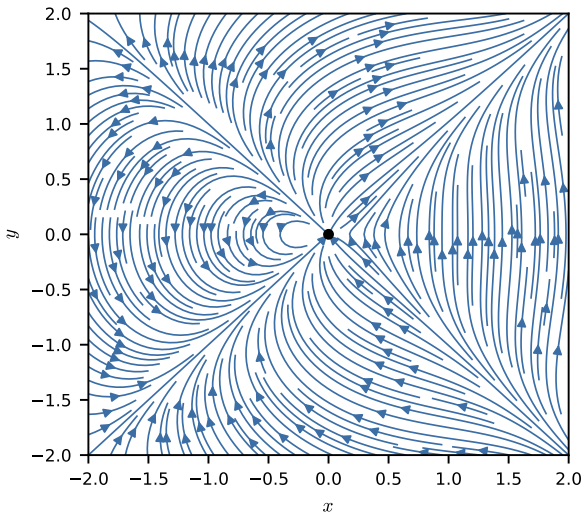

6.3.16: $a = -0.3, 0, 0.3$.

**6.3.17** $\dot{x} = xy - x^2y + y^3,\ \dot{y} = y^2 + x^3 - xy^2$

**Solution.** Polar coordinates: $r\dot{r} = y\, r^2$ and $\dot{\theta} = (x^4 - y^4)/r^2$, giving

$$\dot{r} = r^2 \sin\theta, \qquad \dot{\theta} = r^2 \cos 2\theta.$$

Hence $dr/d\theta = \sin\theta/\cos 2\theta$, independent of $r$: $r = C + F(\theta)$ with $F(\theta) = -\frac{1}{2\sqrt{2}} \ln\left|\frac{\sqrt{2}\cos\theta - 1}{\sqrt{2}\cos\theta + 1}\right|$. The invariant rays are $\theta = \pm\pi/4, \pm 3\pi/4$: $\theta = \pi/4$ (outward, attracts angles), $3\pi/4$ (outward, repels angles), $-\pi/4$ (inward, repels angles), $-3\pi/4$ (inward, attracts angles). The origin is a highly degenerate fixed point (all derivatives of order $\leq 2$ vanish, so no linear information). In the sector $-\pi/4 < \theta < \pi/4$, $r = C + F(\theta) \to \infty$ at both edges, so trajectories come in from infinity near the ray $\theta = -\pi/4$, pass at minimum distance $C + F(0)$ at $\theta = 0$, and leave to infinity along $\theta = \pi/4$: a hyperbolic sector, and the member with $C = -F(0)$ passes through the origin itself. The other sectors, and how these curves approach the origin, are shown in the computer-generated portrait.

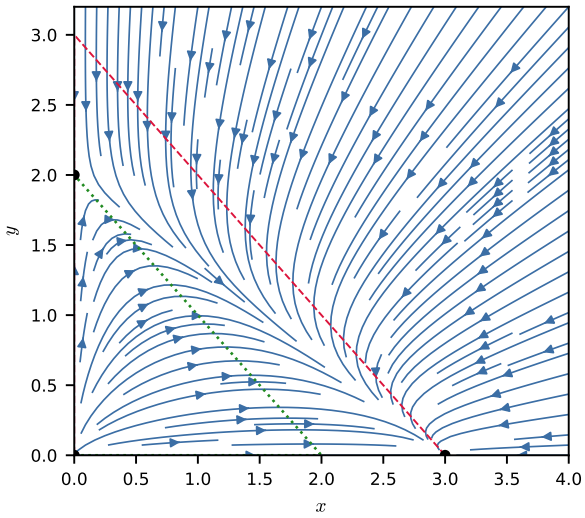



6.3.17.

## 6.4 Rabbits versus Sheep

### 6.4.1 $\dot{x} = x(3 - x - y),\ \dot{y} = y(2 - x - y)$

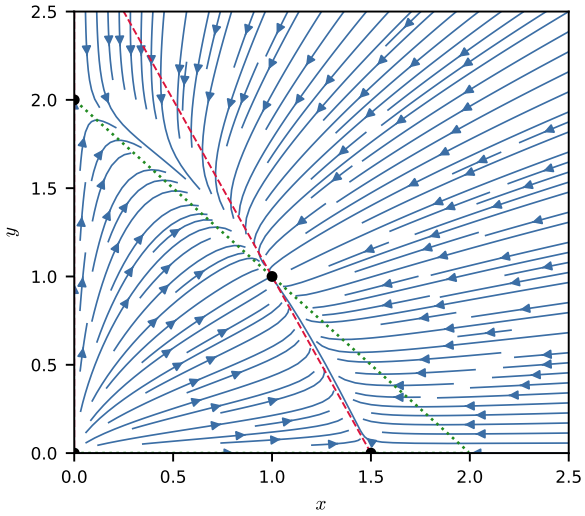
**Solution.** Fixed points: $(0,0), (3,0), (0,2)$; no interior one (the nullclines $x + y = 3$ and $x + y = 2$ are parallel). $A = \begin{pmatrix} 3 - 2x - y & -x \\ -y & 2 - x - 2y \end{pmatrix}$: $(0,0)$: eigenvalues $3, 2$, unstable node. $(3,0)$: $\begin{pmatrix} -3 & -3 \\ 0 & -1 \end{pmatrix}$, eigenvalues $-3, -1$: stable node. $(0,2)$: $\begin{pmatrix} 1 & 0 \\ -2 & -2 \end{pmatrix}$, eigenvalues $1, -2$: saddle. The $y$-axis is the stable manifold of the saddle. **Basin of** $(3,0)$**:** all $x_0 > 0$, $y_0 \geq 0$ (the rabbits always win).




6.4.1.

### 6.4.2 $\dot{x} = x(3 - 2x - y),\ \dot{y} = y(2 - x - y)$

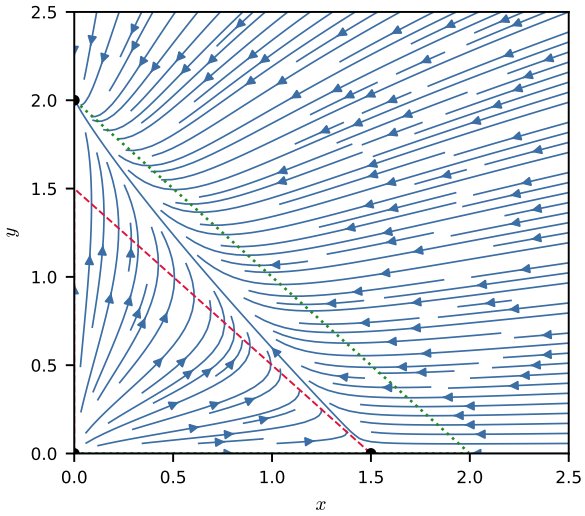
**Solution.** Fixed points: $(0,0), (3/2,0), (0,2), (1,1)$. $A = \begin{pmatrix} 3-4x-y & -x \\ -y & 2-x-2y \end{pmatrix}$: $(0,0)$ unstable node; $(3/2,0)$: eigenvalues $-3, \frac{1}{2}$ saddle; $(0,2)$: eigenvalues $1, -2$ saddle; $(1,1)$: $\begin{pmatrix} -2 & -1 \\ -1 & -1 \end{pmatrix}$, $\tau = -3, \Delta = 1$, discriminant $5 > 0$: stable node. The axes are the stable manifolds of the saddles, so the **basin of** $(1,1)$ **is the open first quadrant** $x > 0, y > 0$: coexistence.




6.4.2.

### 6.4.3 $\dot{x} = x(3 - 2x - 2y),\ \dot{y} = y(2 - x - y)$

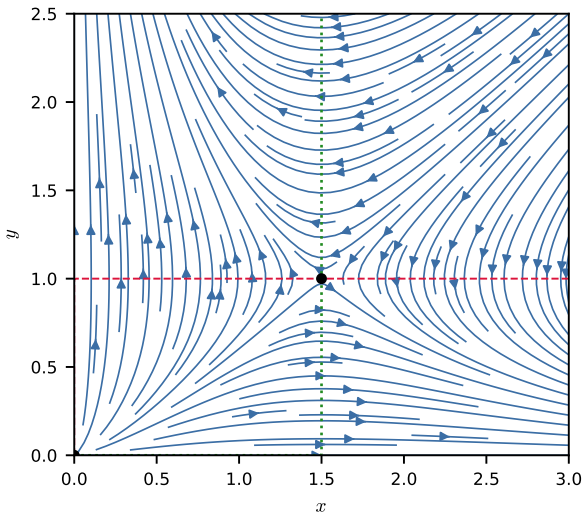
**Solution.** Fixed points $(0,0), (3/2,0), (0,2)$ (nullclines $x+y=3/2$ and $x+y=2$ are parallel). $(0,0)$: unstable node. $(3/2,0)$: $\begin{pmatrix} -3 & -3 \\ 0 & 1/2 \end{pmatrix}$ saddle. $(0,2)$: $\begin{pmatrix} -1 & 0 \\ -2 & -2 \end{pmatrix}$, eigenvalues $-1,-2$: stable node. The $x$-axis is the stable manifold of the saddle, so the **basin of** $(0,2)$ **is** $\{y_0 > 0\}$: the sheep always win.




6.4.3.

### 6.4.4 *Simplest competition model* $\dot{N}_1 = r_1 N_1 - b_1 N_1 N_2,\ \dot{N}_2 = r_2 N_2 - b_2 N_1 N_2$

**Solution.** (a) Less realistic: each species grows exponentially without bound in the absence of the other (no carrying capacity, no self-limitation). (b) Let $x = b_2 N_1/r_1,\ y = b_1 N_2/r_1,\ \tau = r_1 t$. Then $x' = x(1-y)$ and $y' = y(S-x)$ with $\boxed{S = r_2/r_1}$. (c) Nullclines: $x'=0$: $x=0$ or $y=1$; $y'=0$: $y=0$ or $x=S$. Fixed points: $(0,0)$ and $(S,1)$. Signs: $x'>0, y'>0$ for $x<S, y<1$; $x'>0, y'<0$ for $x>S, y<1$; $x'<0, y'<0$ for $x>S, y>1$; $x'<0, y'>0$ for $x<S, y>1$. (d) $A = \begin{pmatrix} 1-y & -x \\ -y & S-x \end{pmatrix}$: $(0,0)$ is an unstable node (eigenvalues $1, S$); at $(S,1)$, $A = \begin{pmatrix} 0 & -S \\ -1 & 0 \end{pmatrix}$, $\lambda = \pm\sqrt{S}$: saddle. Biological meaning: coexistence is unstable. Except for initial conditions exactly on the stable manifold of the saddle, one species grows without bound while the other dies out, depending on initial conditions (and both blow up exponentially, which is unrealistic). (e) $\dfrac{dx}{dy} = \dfrac{x(1-y)}{y(S-x)} \Rightarrow \dfrac{S-x}{x}dx = \dfrac{1-y}{y}dy \Rightarrow S\ln x - x = \ln y - y + C$. The trajectories *not* of this form are those on the coordinate axes ($x=0$ or $y=0$), where the logarithms are undefined, plus the origin.




6.4.4 with $S = 1.5$.

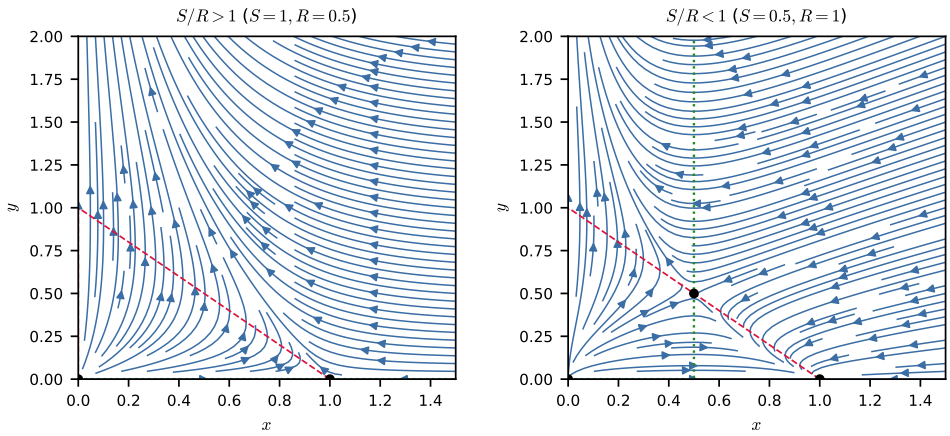
### 6.4.5 *Species 1 with finite carrying capacity $K_1$*

**Solution.** $\dot{N}_1 = r_1 N_1(1 - N_1/K_1) - b_1 N_1 N_2$, $\dot{N}_2 = r_2 N_2 - b_2 N_1 N_2$. Let $x = N_1/K_1$, $y = b_1 N_2/r_1$, $\tau = r_1 t$:

$$x' = x(1 - x - y), \qquad y' = y(S - Rx), \qquad S = \frac{r_2}{r_1},\ R = \frac{b_2 K_1}{r_1}.$$

Nullclines: $x = 0$ or $x + y = 1$; $y = 0$ or $x = S/R$ (the ratio $S/R = r_2/(b_2 K_1)$ is what $K_1$ controls). There is no fixed point on the $y$-axis except the origin (species 2 grows exponentially alone). Fixed points: $(0, 0)$ (unstable node), $(1, 0)$, and the interior point $(S/R,\ 1 - S/R)$ which exists iff $S/R < 1$, i.e. $K_1 > r_2/b_2$. At $(1, 0)$: $A = \begin{pmatrix} -1 & -1 \\ 0 & S - R \end{pmatrix}$.

- $K_1 < r_2/b_2$ **$(S/R > 1)$.** No interior fixed point; $(1, 0)$ is a saddle. Every trajectory with $y_0 > 0$ has $y \to \infty$, $x \to 0$: species 2 explodes and species 1 goes extinct. (Only if $y_0 = 0$ does $x \to 1$.)
- $K_1 > r_2/b_2$ **$(S/R < 1)$.** $(1, 0)$ is a stable node (eigenvalues $-1$, $S - R < 0$) and the interior point is a saddle ($\tau = -x^*$, $\Delta = -Rx^* y^* < 0$). The stable manifold of the saddle splits the quadrant: below it species 1 wins ($x \to 1, y \to 0$); above it species 2 grows without bound while species 1 dies out. Bistability.




6.4.5: the two qualitatively different portraits.

### 6.4.6 *Both species with finite carrying capacity*

**Solution.** (a) With $x = N_1/K_1,\ y = N_2/K_2,\ \tau = r_1 t$:

$$x' = x(1 - x - ay), \qquad y' = \rho\, y(1 - y - bx), \qquad \rho = \frac{r_2}{r_1},\ a = \frac{b_1 K_2}{r_1},\ b = \frac{b_2 K_1}{r_2}.$$

**Three** dimensionless groups $(\rho, a, b)$: six dimensional parameters minus three independent units (time, $N_1$, $N_2$). ($\rho$ only changes the relative speed of $y$, not the topology.) (b) Nullclines: $L_1$ : $x + ay = 1$ (intercepts $(1,0), (0,1/a)$) and $L_2$ : $bx + y = 1$ (intercepts $(1/b, 0), (0,1)$). Fixed points $(0,0), (1,0), (0,1)$ plus an interior point iff $L_1, L_2$ cross in the quadrant. The four configurations:

(i) $a < 1, b < 1$: nullclines cross, interior point is a stable node (coexistence).

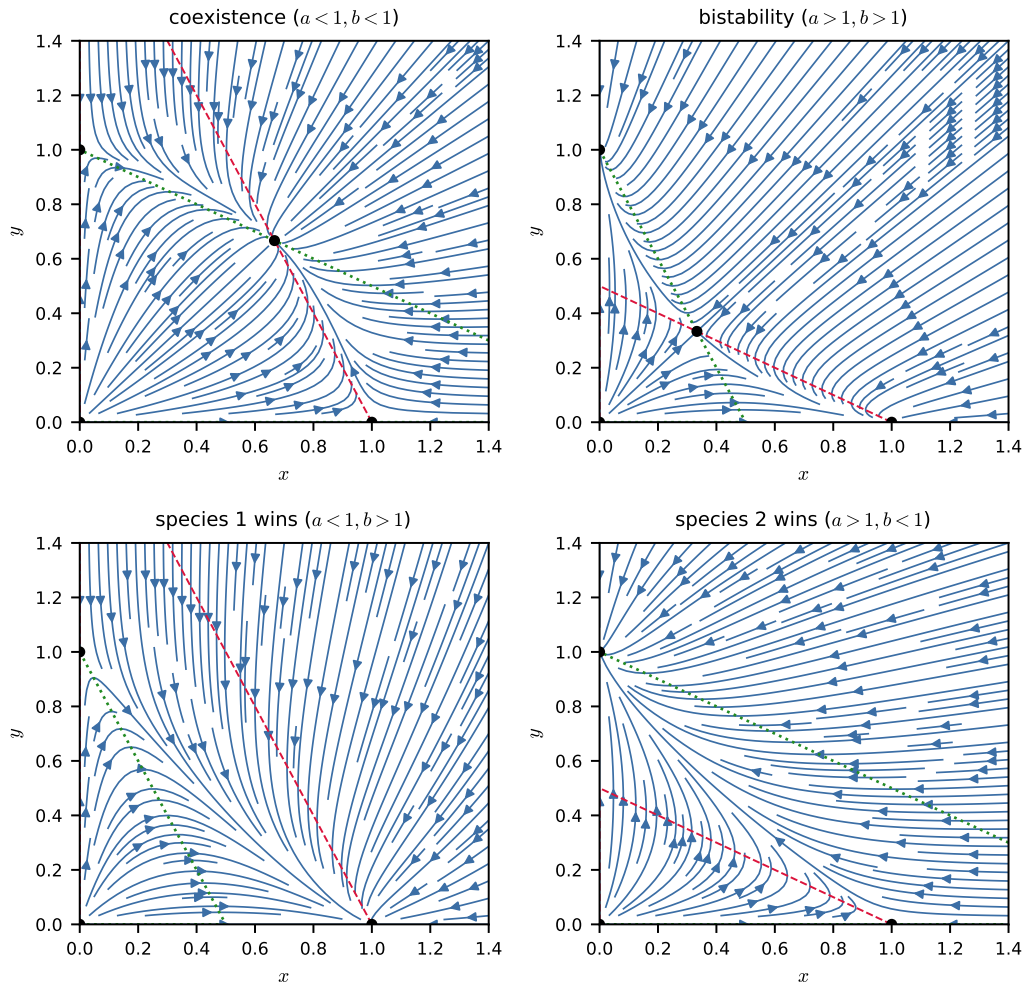
(ii) $a > 1, b > 1$: nullclines cross, interior point is a saddle; $(1,0)$ and $(0,1)$ both stable (bistability).

(iii) $a < 1, b > 1$: $L_1$ lies above $L_2$, no crossing; $(1,0)$ stable, $(0,1)$ saddle (species 1 wins).

(iv) $a > 1, b < 1$: $L_1$ below $L_2$; $(0,1)$ stable, $(1,0)$ saddle (species 2 wins).

Stability: $A(1,0) = \begin{pmatrix} -1 & -a \\ 0 & \rho(1-b) \end{pmatrix}$ stable iff $b > 1$; $A(0,1) = \begin{pmatrix} 1-a & 0 \\ -\rho b & -\rho \end{pmatrix}$ stable iff $a > 1$.

(c) Interior point $x^* = \dfrac{1-a}{1-ab},\ y^* = \dfrac{1-b}{1-ab}$. $A = \begin{pmatrix} -x^* & -ax^* \\ -\rho b y^* & -\rho y^* \end{pmatrix}$, $\tau = -(x^* + \rho y^*)$, $\Delta = \rho x^* y^* (1 - ab)$. For $a, b < 1$: $x^*, y^* > 0$, $\Delta > 0$, $\tau < 0$, and $\tau^2 - 4\Delta = (x^* - \rho y^*)^2 + 4\rho a b x^* y^* > 0$: stable node. For $a, b > 1$: $\Delta < 0$, saddle. **Stable coexistence iff** $a < 1$ **and** $b < 1$, i.e. $b_1 K_2 < r_1$ and $b_2 K_1 < r_2$: the competitive effect of a full population of the other species on a species is weaker than that species' own self-limitation (inter-specific competition weaker than intra-specific).

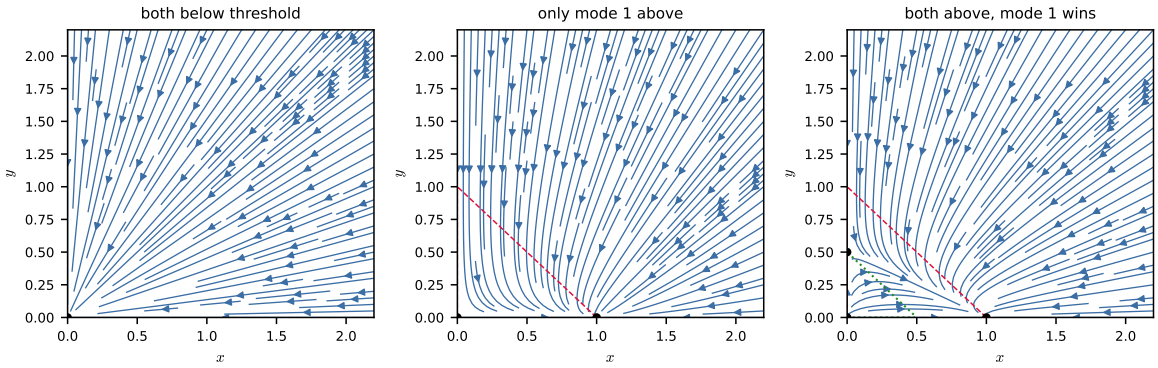



6.4.6: coexistence, bistability, and the two exclusion cases.

### 6.4.7 *Two-mode laser*

**Solution.** $\dot{n}_i = G_i N n_i - k_i n_i$, $N = N_0 - B_1 n_1 - B_2 n_2$. (a) At the origin, $A = \mathrm{diag}(G_1 N_0 - k_1,\ G_2 N_0 - k_2)$. It is stable iff $G_1 N_0 < k_1$ and $G_2 N_0 < k_2$ (both modes below threshold); if one inequality is reversed it is a saddle, if both a source. (b) $n_2 = 0, n_1 \neq 0$: $G_1 N = k_1$, so $n_1^* = (G_1 N_0 - k_1)/(G_1 B_1)$, which is $> 0$ iff $G_1 N_0 > k_1$. Similarly $n_2^* = (G_2 N_0 - k_2)/(G_2 B_2)$. A fixed point with both $n_i > 0$ would need $G_1 N = k_1$ and $G_2 N = k_2$ simultaneously: only possible in the non-generic case $k_1/G_1 = k_2/G_2$ (then a whole line of fixed points). At $(n_1^*, 0)$: $A = \begin{pmatrix} -G_1 B_1 n_1^* & -G_1 B_2 n_1^* \\ 0 & G_2 k_1/G_1 - k_2 \end{pmatrix}$, eigenvalues $-G_1 B_1 n_1^* < 0$ and $G_2 k_1/G_1 - k_2$. So $(n_1^*, 0)$ is a stable node iff $k_1/G_1 < k_2/G_2$ (mode 1 has the lower threshold), otherwise a saddle. Symmetrically for $(0, n_2^*)$. (c) Generic portraits: (i) both below threshold: only the origin (stable): laser off; (ii) only mode 1 above threshold: origin a saddle, $(n_1^*, 0)$ a stable node; (iii) the mirror image; (iv) both above threshold: origin a source, the mode with the lower threshold $k_i/G_i$ is a stable node and the other axis fixed point a saddle. That is five portraits (three up to relabeling modes). Prediction: the laser operates in at most one mode, the one with the lowest threshold: winner-take-all mode

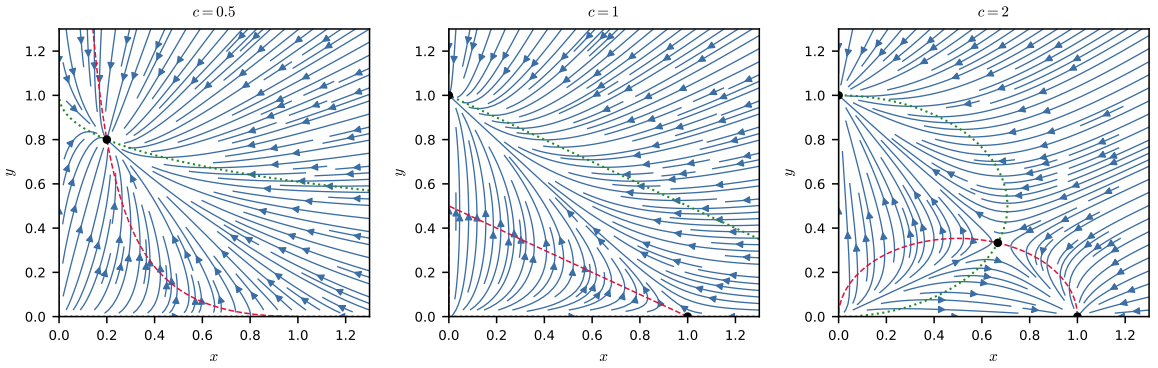
competition.




6.4.7.

### 6.4.8 $\dot{x} = ax^c - Gx,\ \dot{y} = by^c - Gy,\ G = ax^c + by^c$

**Solution.** (a) $s = x + y$: $\dot{s} = G - Gs = G(1 - s)$. Along any solution $G(t)$ is a known function, so $s$ solves the linear equation $\dot{s} = G(t)(1 - s)$; $s(0) = 1$ gives $s \equiv 1$ by uniqueness. (b) With $u = 1 - s$, $\dot{u} = -Gu$. If $s_0 \in (0, \infty)$, $u$ keeps its sign and $|u|$ decreases, so $s$ stays between $s_0$ and 1, hence bounded away from 0 and $G \geq \kappa s^c \geq G_{\min} > 0$ (using $x^c + y^c \geq \min(1, 2^{1-c})s^c$). Then $|u| \leq |u_0|e^{-G_{\min}t} \to 0$: all trajectories in the positive quadrant approach the line $x + y = 1$. (c) $c = 1$: on $y = 1 - x$, $G = b + (a - b)x$ and $\dot{x} = (a - b)x(1 - x)$. If $a > b$ then $x \to 1$ (species $x$ wins); if $a < b$, $x \to 0$ ($y$ wins); if $a = b$ the whole line is fixed. (d) General $c$: on the line, $\dot{x} = x(1 - x)\big[ax^{c-1} - b(1 - x)^{c-1}\big]$. For $c > 1$ the bracket is $\approx -b < 0$ near $x = 0$ and $\approx a > 0$ near $x = 1$, so both end points $(0, 1), (1, 0)$ are stable and the interior fixed point $x^*/(1 - x^*) = (b/a)^{1/(c-1)}$ is unstable: bistability (winner depends on initial condition). (e) $c < 1$: now $x^{c-1} \to \infty$ at $x = 0$, so $\dot{x} > 0$ near $x = 0$ and $< 0$ near $x = 1$: the end points are unstable and the interior point $x^*/(1 - x^*) = (a/b)^{1/(1-c)}$ is stable: coexistence.



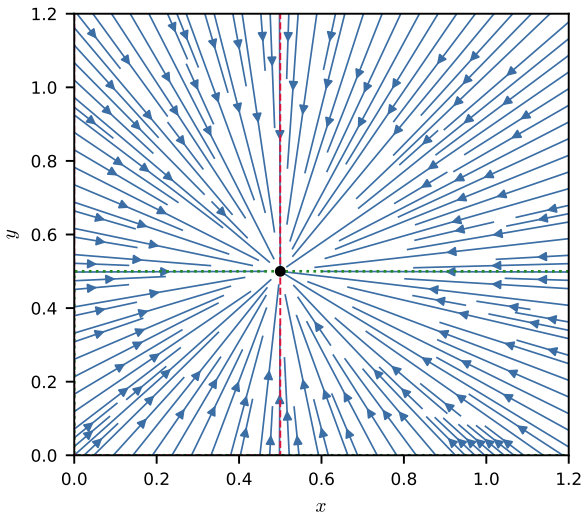

6.4.8 with $a = 1, b = 2$.

### 6.4.9 *Model of a national economy:* $\dot{I} = I - \alpha C,\ \dot{C} = \beta(I - C - G)$

**Solution.** (Note: the Greek letters $\alpha, \beta$ appear as $B, C$ in some copies of the text; the printed conditions are $1 < \alpha < \infty$ and $1 \le \beta < \infty$.) (a) $G$ constant: $I = \alpha C$, $I - C = G \Rightarrow C^* = G/(\alpha - 1)$, $I^* = \alpha G/(\alpha - 1)$. $A = \begin{pmatrix} 1 & -\alpha \\ \beta & -\beta \end{pmatrix}$, $\tau = 1 - \beta$, $\Delta = \beta(\alpha - 1) > 0$. For $\beta > 1$: stable ($\tau < 0$), a spiral if $(\beta-1)^2 < 4\beta(\alpha-1)$, a node otherwise. For $\beta = 1$: $\tau = 0, \Delta = \alpha - 1 > 0$: a center (the system is linear), so the economy oscillates. (b) $G = G_0 + kI$: $I = \alpha C$ and $(1-k)I - C = G_0$ give $C^* = \dfrac{G_0}{\alpha(1-k) - 1}$, $I^* = \alpha C^*$. This is in the first quadrant iff $\alpha(1-k) > 1$, i.e. $k < k_c = 1 - 1/\alpha$. At $k = k_c$ the equilibrium runs off to infinity. Also $\Delta = \beta(\alpha(1-k) - 1)$ changes sign at $k_c$. For $k > k_c$ there is no sensible equilibrium; the only fixed point is a saddle in the third quadrant and trajectories in the first quadrant run away: income and consumption grow without bound (runaway expansion). (c) $G = G_0 + kI^2$: fixed points satisfy $kI^2 - (1 - 1/\alpha)I + G_0 = 0$, $C = I/\alpha$. With $G_c = (1 - 1/\alpha)^2/(4k)$: two positive roots for $G_0 < G_c$, one for $G_0 = G_c$, none for $G_0 > G_c$. At a root, $\Delta = \beta[\alpha(1 - 2kI) - 1]$ with $2kI_\pm = (1 - 1/\alpha) \pm \sqrt{D}$, $D = (1 - 1/\alpha)^2 - 4kG_0$: at $I_-$, $\Delta = \beta\alpha\sqrt{D} > 0$ (stable for $\beta > 1$, center for $\beta = 1$); at $I_+$, $\Delta = -\beta\alpha\sqrt{D} < 0$ (saddle). So for small government spending base $G_0$ there is a healthy stable equilibrium (with a finite basin bounded by the saddle's stable manifold; states outside run away), the two merge in a saddle-node at $G_0 = G_c$, and for $G_0 > G_c$ there is no equilibrium: the economy has no steady state.

### 6.4.10 *Hypercycle ($n = 2$)*

**Solution.** (a) With $x_0 = x_2$: $\sum_j x_j x_{j-1} = x_1 x_0 + x_2 x_1 = 2x_1 x_2$, so $\dot{x}_1 = x_1(x_2 - 2x_1 x_2)$ and $\dot{x}_2 = x_2(x_1 - 2x_1 x_2)$. (b) For $x_i > 0$ the only fixed point solves $1 - 2x_1 = 0$, $1 - 2x_2 = 0$: $(\frac{1}{2}, \frac{1}{2})$. There $A = \begin{pmatrix} x_2 - 4x_1x_2 & x_1 - 2x_1^2 \\ x_2 - 2x_2^2 & x_1 - 4x_1x_2 \end{pmatrix} = -\frac{1}{2}I$: stable star. (On the boundary the axes are lines of non-isolated fixed points $(p, 0)$ with eigenvalues $0, p$, repelling transversally.) (c) $u = x_1 + x_2$: $\dot{u} = 2x_1x_2(1-u)$. Also $v = x_1 - x_2$: $\dot{v} = -2x_1x_2 v$. Let $d\sigma = 2x_1x_2\,dt$. Then $du/d\sigma = 1-u$, $dv/d\sigma = -v$, so $u - 1 = (u_0 - 1)e^{-\sigma}$ and $v = v_0 e^{-\sigma}$. Hence $x_1 = \frac{1}{2} + (x_1(0) - \frac{1}{2})e^{-\sigma}$, $x_2 = \frac{1}{2} + (x_2(0) - \frac{1}{2})e^{-\sigma}$ are convex combinations of $x_i(0)$ and $\frac{1}{2}$, so they stay $\ge \min(x_i(0), \frac{1}{2}) > 0$. Therefore $2x_1x_2 \ge c > 0$ and $\sigma(t) \ge ct \to \infty$. Thus $u \to 1$ as $t \to \infty$. (d) Likewise $v = v_0 e^{-\sigma(t)} \to 0$. (e) $u \to 1, v \to 0$ gives $(x_1, x_2) \to (\frac{1}{2}, \frac{1}{2})$. (f) The computer portrait shows all trajectories are *straight lines* converging to $(\frac{1}{2}, \frac{1}{2})$. Explanation: $v/(u-1) = v_0/(u_0 - 1)$ is constant, i.e. $x_1 - x_2 = c\,(x_1 + x_2 - 1)$, a line through $(\frac{1}{2}, \frac{1}{2})$.




6.4.10: straight-line trajectories into $(\frac{1}{2}, \frac{1}{2})$.

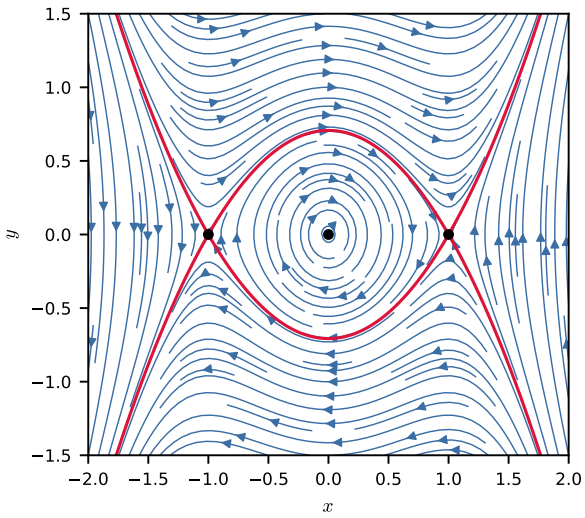
### 6.4.11 *Leftists, rightists, centrists*

**Solution.** (a) $(x+y+z)^{\cdot} = rxz+ryz-rxz-ryz = 0$, so $x+y+z$ is constant; the set $x+y+z=1$ is invariant. (b) On it $\dot{z} = -rz(x+y) = -rz(1-z)$. Also $(x/y)^{\cdot} = 0$, so the ratio of extremists is constant. Fixed points: $z=0$ (any $x+y=1$) and $(0,0,1)$. $r>0$*:* $z \to 0$ (the centrist state is unstable), and $x \to x_0/(x_0+y_0)$, $y \to y_0/(x_0+y_0)$: the centrists disappear and the population ends polarized in the initial ratio of rightists to leftists. $r<0$*:* $z \to 1$ and $x, y \to 0$: everyone ends up a centrist (consensus in the middle). (c) If extremists are persuasive ($r>0$), the middle is hollowed out and opinion polarizes; if centrists are persuasive ($r<0$), extremism dies out.

## 6.5 Conservative Systems

### 6.5.1 $\ddot{x} = x^3 - x$

**Solution.** $\dot{x} = y$, $\dot{y} = x^3 - x$; potential $V = \frac{1}{2}x^2 - \frac{1}{4}x^4$ ($\ddot{x} = -V'$). (a) Equilibria $x = 0, \pm 1$; $A = \begin{pmatrix} 0 & 1 \\ 3x^2-1 & 0 \end{pmatrix}$. At 0: $\Delta = 1$, center (a nonlinear center, since $V$ has an isolated minimum). At $\pm 1$: $\Delta = -2$, saddles. (b) $E = \frac{1}{2}y^2 + \frac{1}{2}x^2 - \frac{1}{4}x^4$. (c) Closed orbits around the origin for $0 < E < \frac{1}{4}$; at $E = \frac{1}{4}$ two heteroclinic orbits $y = \pm(1-x^2)/\sqrt{2}$ connect the saddles; the remaining orbits are unbounded.

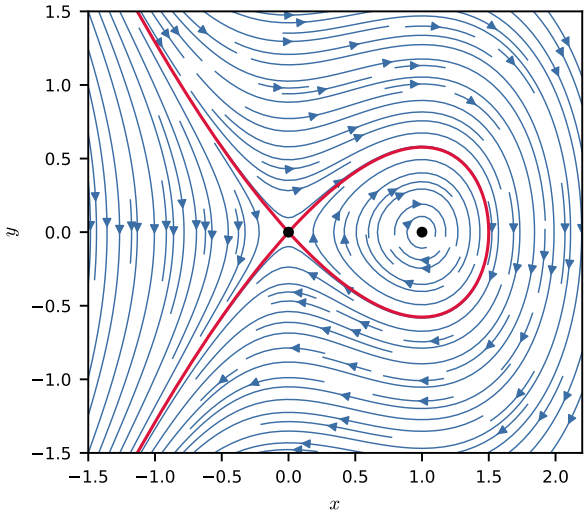



6.5.1 (red: saddle energy $E = \frac{1}{4}$).

### 6.5.2 $\ddot{x} = x - x^2$

**Solution.** $V = -\frac{1}{2}x^2 + \frac{1}{3}x^3$. (a) $A = \begin{pmatrix} 0 & 1 \\ 1-2x & 0 \end{pmatrix}$: $x = 0$: $\Delta = -1$, saddle; $x = 1$: $\Delta = 1$, nonlinear center ($V(1) = -\frac{1}{6}$ is a local minimum). (b) Closed orbits surround $(1, 0)$, bounded by a homoclinic loop through the saddle; all other orbits are unbounded. (c) Homoclinic orbit: $E = V(0) = 0$, so $\frac{1}{2}y^2 = \frac{1}{2}x^2 - \frac{1}{3}x^3$:

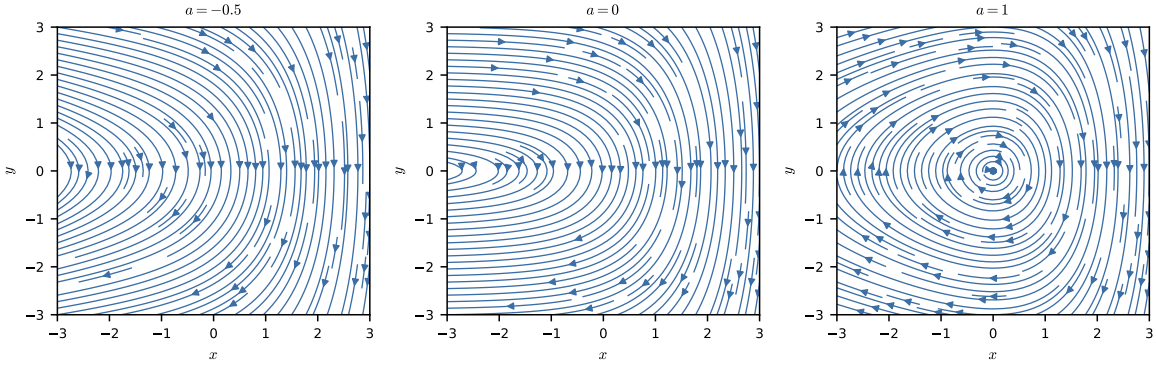
$$y = \pm x\sqrt{1 - \tfrac{2}{3}x}, \qquad 0 \le x \le \tfrac{3}{2}.$$




6.5.2.

### 6.5.3 $\ddot{x} = a - e^x$

**Solution.** $V = e^x - ax$, $E = \frac{1}{2}y^2 + e^x - ax$ is conserved. $a > 0$*:* equilibrium $x^* = \ln a$ with $\Delta = e^{x^*} = a > 0$: a center. $V \to +\infty$ as $x \to \pm\infty$, so *all* orbits are closed. $a = 0$*:* no equilibria; $E = \frac{1}{2}y^2 + e^x$: U-shaped orbits opening toward $x \to -\infty$, where $y \to \pm\sqrt{2E}$. $a < 0$*:* no equilibria; $V \to -\infty$ as $x \to -\infty$; U-shaped orbits whose arms have $|y| \to \infty$ as $x \to -\infty$.



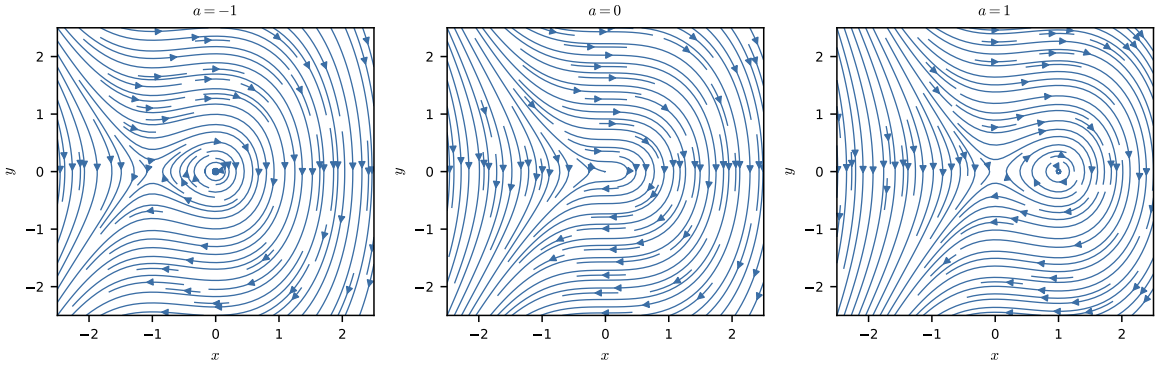

6.5.3: $a < 0$, $a = 0$, $a > 0$.

### 6.5.4 $\ddot{x} = ax - x^2$

**Solution.** Equilibria $x = 0$ and $x = a$. $A = \begin{pmatrix} 0 & 1 \\ a - 2x & 0 \end{pmatrix}$, $\Delta = 2x - a$. At $x = 0$: $\Delta = -a$; at $x = a$: $\Delta = a$.

- $a < 0$: $x = a$ is a saddle, $x = 0$ a nonlinear center; a homoclinic loop through the saddle, $y^2 = -\frac{1}{3}(x - a)^2(2x + a)$ for $a \le x \le -a/2$, encloses the center.
- $a = 0$: $\ddot{x} = -x^2$ has a single (degenerate) equilibrium; $E = \frac{1}{2}y^2 + \frac{1}{3}x^3$: orbits are U-shaped curves opening to the left, and $E = 0$ is the cuspidal curve $y^2 = -\frac{2}{3}x^3$.
- $a > 0$: $x = 0$ is a saddle, $x = a$ a center; homoclinic loop $y = \pm x\sqrt{a - \frac{2}{3}x}$, $0 \le x \le \frac{3}{2}a$, around the center.

(Transcritical-type exchange of stability at $a = 0$.)




6.5.4: $a = -1, 0, 1$.

### 6.5.5 $\ddot{x} = (x-a)(x^2-a)$

**Solution.** $F(x) = (x-a)(x^2-a)$, $\ddot{x} = F(x) = -V'$. Equilibria: $x = a$, and $x = \pm\sqrt{a}$ if $a > 0$. Each has $A = \begin{pmatrix} 0 & 1 \\ F' & 0 \end{pmatrix}$, $\Delta = -F'(x^*)$: saddle if $F' > 0$, (nonlinear) center if $F' < 0$ (conservative system), degenerate if $F' = 0$. $F$ is a cubic with positive leading coefficient, so at consecutive simple roots $F'$ alternates in sign starting with $+$ at the smallest.

- $a < 0$: only the root $x = a$; $F'(a) = a^2 - a > 0$: saddle.
- $a = 0$: $\ddot{x} = x^3$, triple root at 0; $V = -x^4/4$ has a maximum: unstable, saddle-like (degenerate).
- $0 < a < 1$: roots $-\sqrt{a} < a < \sqrt{a}$: saddle at $-\sqrt{a}$, center at $a$, saddle at $\sqrt{a}$.
- $a = 1$: roots $-1$ and double root 1: saddle at $-1$; $x = 1$ is a degenerate (cusp-like) point since $F = (x-1)^2(x+1) \geq 0$ near it.
- $a > 1$: roots $-\sqrt{a} < \sqrt{a} < a$: saddle at $-\sqrt{a}$, center at $\sqrt{a}$, saddle at $a$.

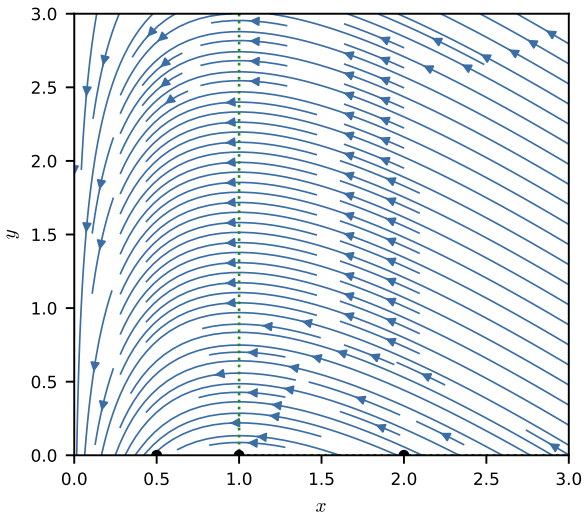
So there is a pitchfork-like bifurcation at $a = 0$ and an exchange of stability between $x = a$ and $x = \sqrt{a}$ at $a = 1$.

### 6.5.6 *Epidemic model:* $\dot{x} = -kxy,\ \dot{y} = kxy - \ell y$

**Solution.** (a) $\dot{x} = 0 \Rightarrow x = 0$ or $y = 0$; then $\dot{y} = 0$ forces $y = 0$. Fixed points: the whole $x$-axis $(x^*, 0)$ (non-isolated). $A = \begin{pmatrix} -ky & -kx \\ ky & kx - \ell \end{pmatrix} \to \begin{pmatrix} 0 & -kx^* \\ 0 & kx^* - \ell \end{pmatrix}$, eigenvalues 0 and $kx^* - \ell$: stable (transversally) if $x^* < \ell/k$, unstable if $x^* > \ell/k$. (b) Nullclines: $\dot{x} = 0$ on both axes; $\dot{y} = 0$ on $y = 0$ and $x = \ell/k$. For $x > \ell/k, y > 0$: $\dot{x} < 0, \dot{y} > 0$; for $0 < x < \ell/k$: $\dot{x} < 0, \dot{y} < 0$. (c) $dy/dx = -1 + \ell/(kx)$, so $y = -x + \frac{\ell}{k}\ln x + C$: conserved quantity

$$x + y - \tfrac{\ell}{k}\ln x = C.$$

(d) Orbits are the concave curves $y = C - x + \frac{\ell}{k}\ln x$ with maximum at $x = \ell/k$. Since $\dot{x} < 0$ the flow runs leftward: from the axis at some $x > \ell/k$, up over the maximum, down to the axis at $x_\infty < \ell/k$. As $t \to \infty$: $y \to 0$, $x \to x_\infty < \ell/k$ (the epidemic burns out, with some healthy people left). (e) An epidemic occurs iff $\dot{y}(0) > 0$, i.e. $y_0(kx_0 - \ell) > 0$: $x_0 > \ell/k$.




6.5.6 ($k = \ell = 1$).

### 6.5.7 *Relativistic orbit:* $u'' + u = \alpha + \varepsilon u^2$

**Solution.** (a) $u' = v,\ v' = \alpha - u + \varepsilon u^2$. (b) Equilibria: $v = 0$, $\varepsilon u^2 - u + \alpha = 0$: $u_\pm = \dfrac{1 \pm \sqrt{1 - 4\alpha\varepsilon}}{2\varepsilon}$ (real iff $4\alpha\varepsilon \le 1$), $u_- \approx \alpha + \varepsilon\alpha^2$, $u_+ \approx 1/\varepsilon$. (c) $A = \begin{pmatrix} 0 & 1 \\ -1 + 2\varepsilon u & 0 \end{pmatrix}$, $\Delta = 1 - 2\varepsilon u$. At $u_-$: $\Delta = \sqrt{1 - 4\alpha\varepsilon} > 0$: linear center; at $u_+$: $\Delta < 0$: saddle. The system is conservative with $E = \frac{1}{2}v^2 + \frac{1}{2}u^2 - \alpha u - \frac{1}{3}\varepsilon u^3$, and $u_-$ is a strict local minimum of $E$, so it is a nonlinear center. (d) $u = \text{const}$ means $r = 1/u$ constant: a circular orbit of radius $\approx 1/\alpha$. Nearby closed orbits in the $(u, v)$ plane are slightly eccentric planetary orbits; their frequency in $\theta$ is $\sqrt{\Delta} = (1 - 4\alpha\varepsilon)^{1/4} < 1$, so they do not close after $2\pi$: the perihelion advances by about $4\pi\alpha\varepsilon$ per revolution (Einstein's correction).

### 6.5.8 *Harmonic oscillator*

**Solution.** $H = \dfrac{p^2}{2m} + \dfrac{kx^2}{2}$: $\dot{x} = \partial H/\partial p = p/m$ (definition of momentum), $\dot{p} = -\partial H/\partial x = -kx$ (Hooke's force). Then $m\ddot{x} = \dot{p} = -kx$, i.e. $F = ma$ with $F = -kx$. $H = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$ is kinetic plus potential energy.

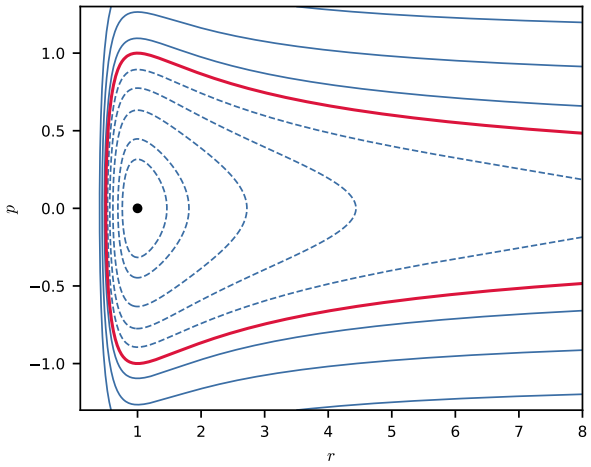
### 6.5.9 *$H$ is conserved*

**Solution.** $\dot{H} = \dfrac{\partial H}{\partial q}\dot{q} + \dfrac{\partial H}{\partial p}\dot{p} = \dfrac{\partial H}{\partial q}\dfrac{\partial H}{\partial p} - \dfrac{\partial H}{\partial p}\dfrac{\partial H}{\partial q} = 0$. So trajectories lie on level curves $H = C$.

### 6.5.10 *Inverse-square law:* $H = \frac{1}{2}p^2 + \dfrac{h^2}{2r^2} - \dfrac{k}{r}$

**Solution.** $\dot{r} = p,\ \dot{p} = h^2/r^3 - k/r^2$. Effective potential $V(r) = h^2/(2r^2) - k/r$. (a) $k > 0$: $V \to +\infty$ as $r \to 0^+$, $V \to 0^-$ as $r \to \infty$, with a single minimum at $r = h^2/k$, $V_{\min} = -k^2/2h^2$. Level curves $\frac{1}{2}p^2 + V(r) = E$: for $E < -k^2/2h^2$ empty; $E = -k^2/2h^2$ the fixed point (circular

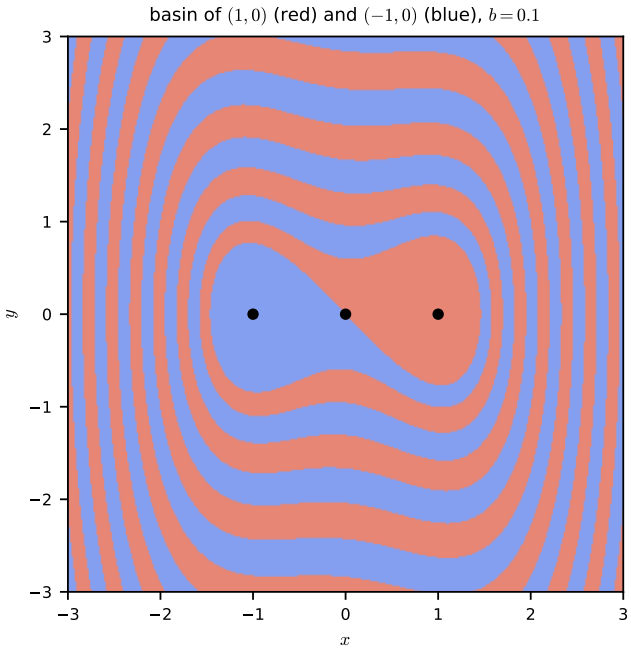
orbit); $-k^2/2h^2 < E < 0$ closed curves around it; $E = 0$ and $E > 0$ unbounded curves reaching $r \to \infty$ ($p \to 0$ for $E = 0$, $p \to \pm\sqrt{2E}$ for $E > 0$). Flow: $\dot{r} = p$, so rightward in $p > 0$. (b) Closed trajectories for $-k^2/2h^2 < E < 0$: the particle is captured (bounded elliptical orbit). $E = 0$: parabolic escape ($r \to \infty$ with speed $\to 0$); $E > 0$: hyperbolic escape with nonzero asymptotic speed. (c) $k < 0$: $V = h^2/(2r^2) + |k|/r$ is monotonically decreasing, no minimum, and $\dot{p} = h^2/r^3 + |k|/r^2 > 0$: no fixed points, hence no closed orbits (a closed orbit would have to enclose one). All orbits scatter: the particle comes in, bounces off the barrier, and leaves.




6.5.10(a): level curves of $H$, $h = k = 1$ (red: $E = 0$).

### 6.5.11 *Basin of attraction for the damped double well*

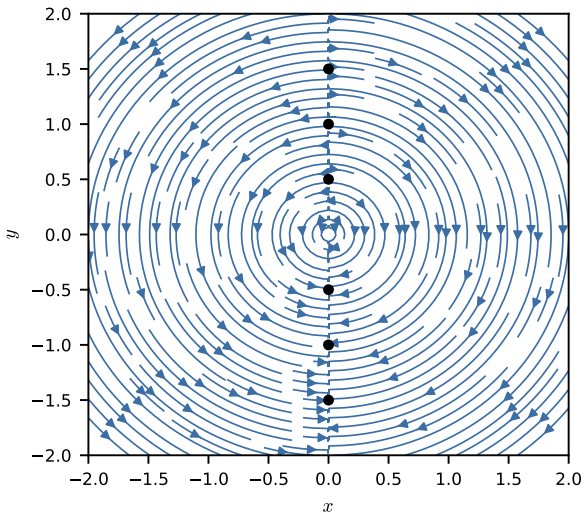
**Solution.** $\dot{x} = y$, $\dot{y} = -by + x - x^3$, $b = 0.1$. Fixed points: saddle at 0, stable spirals at $(\pm 1, 0)$. The energy $E = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4$ obeys $\dot{E} = -by^2 \leq 0$. Once $E < 0$ the particle is trapped in one well, so the basin of $(1, 0)$ contains the part of $\{E < 0\}$ with $x > 0$. At larger energies the particle loses energy slowly and each time it crosses near the barrier it may fall into either well. The result is a pair of interleaved spiral bands that extend to infinity (the stable manifold of the saddle winds around both wells). The plot below shows the basin of $(1, 0)$ in red.




6.5.11, computed by integrating from a $360 \times 360$ grid.

### 6.5.12 $\dot{x} = xy,\ \dot{y} = -x^2$

**Solution.** (a) $\dot{E} = 2x(xy) + 2y(-x^2) = 0$ for $E = x^2 + y^2$. (b) $xy = 0$ and $x^2 = 0$ give $x = 0$: the whole $y$-axis consists of fixed points, so the origin is not isolated. (c) On $E = r^2$: $\dot{y} = -x^2 \leq 0$, so $y$ decreases whenever $x \neq 0$. In the right half-plane $\dot{x} = xy$ has the sign of $y$, and the flow along each circle goes from $(0, r)$ down to $(0, -r)$ (clockwise); in the left half-plane from $(0, r)$ down to $(0, -r)$ counterclockwise. Each orbit is a half circle running from a fixed point $(0, r)$ to the fixed point $(0, -r)$. No closed orbits surround the origin, even though $E$ has a minimum there; the theorem requires an isolated fixed point. At $(0, y_0)$ the eigenvalues are $y_0$ and 0, so points with $y_0 > 0$ repel and those with $y_0 < 0$ attract transversally.



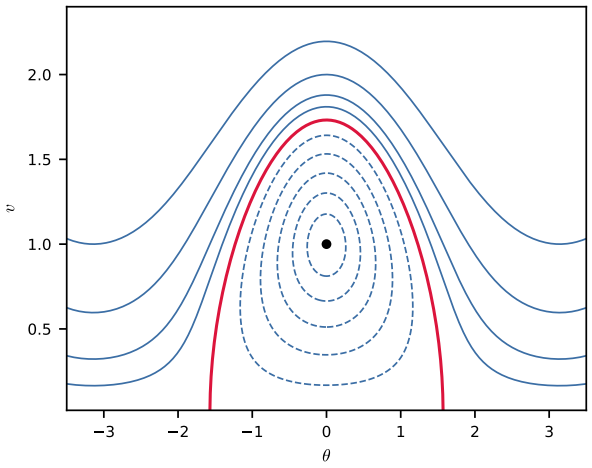

6.5.12.

### 6.5.13 *Nonlinear centers:* $\ddot{x} + x + \varepsilon x^3 = 0$

**Solution.** (a) $\varepsilon > 0$: $V = \frac{1}{2}x^2 + \frac{1}{4}\varepsilon x^4$ has a unique, isolated global minimum at 0 and $V \to \infty$, so every level curve $E = \frac{1}{2}y^2 + V$ is a closed curve: the origin is a nonlinear center (and all orbits are closed). (b) $\varepsilon < 0$: $V = \frac{1}{2}x^2 - \frac{1}{4}|\varepsilon|x^4$ has a local minimum at 0 and maxima at $x = \pm 1/\sqrt{|\varepsilon|}$ with $V = 1/(4|\varepsilon|)$. For $E < 1/(4|\varepsilon|)$ orbits near the origin are closed. Orbits with $E > 1/(4|\varepsilon|)$, or starting beyond the maxima, are unbounded (they escape to infinity in finite time); $E = 1/(4|\varepsilon|)$ gives heteroclinic orbits joining the saddles $(\pm 1/\sqrt{|\varepsilon|}, 0)$.

### 6.5.14 *Glider:* $\dot{v} = -\sin\theta - Dv^2,\ v\dot{\theta} = -\cos\theta + v^2$

**Solution.** (a) $D = 0$: for $E = v^3 - 3v\cos\theta$, $\dot{E} = 3(v^2 - \cos\theta)\dot{v} + 3v\sin\theta\,\dot{\theta} = -3\sin\theta(v^2 - \cos\theta) + 3\sin\theta(v^2 - \cos\theta) = 0$. The only fixed point (with $v > 0$) is $(v, \theta) = (1, 0)$ with $A = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$ (in the order $(v, \theta)$): $\Delta = 2 > 0$, a center; $E$ has a strict minimum $-2$ there (Hessian $\mathrm{diag}(6, 3)$), so it is a nonlinear center. Physically: the glider executes periodic oscillations in speed and flight angle ("phugoid" oscillations): it dives, speeds up, climbs, slows, and so on, tracing a wavy flight path. For $E > 0$ the orbits have $\theta$ increasing without bound: the glider loops the loop; near $v = 0$ (stall) the equations are singular. (b) $D > 0$: $\dot{E} = -3Dv^2(v^2 - \cos\theta)$. Fixed point: $\tan\theta^* = -D$, $v^{*2} = \cos\theta^* = (1 + D^2)^{-1/2}$: a steady glide at a downward angle. At it $A = \begin{pmatrix} -2Dv & -v^2 \\ 2 & -Dv \end{pmatrix}$ (using $\sin\theta = -Dv^2, \cos\theta = v^2$), $\tau = -3Dv^* < 0$, $\Delta = 2v^{*2}(1 + D^2) > 0$: stable; a spiral if $D^2 < 8$ (damped phugoid oscillations decaying to the steady glide), a node if $D^2 > 8$.



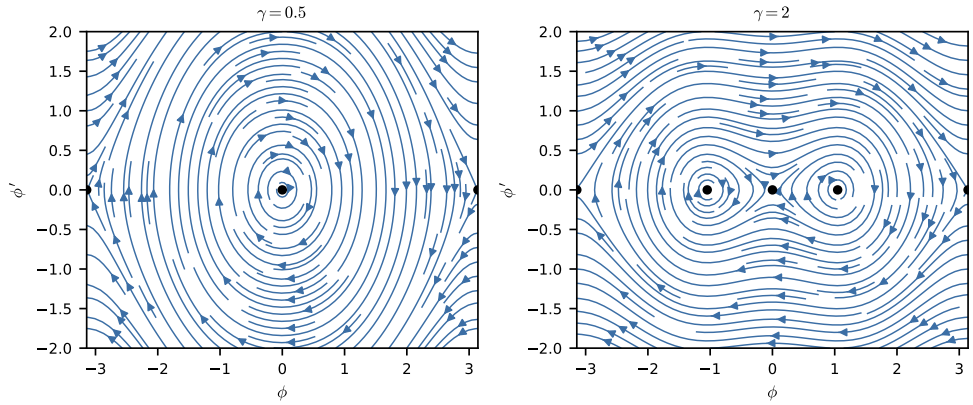

6.5.14(a): level curves of $v^3 - 3v\cos\theta$ (red: $E = 0$).

### 6.5.15 *Frictionless bead on a rotating hoop*

**Solution.** (a) $b = 0$: $mr\ddot{\phi} = -mg\sin\phi + mr\omega^2\sin\phi\cos\phi$. With $\tau = \omega t$, $\gamma = r\omega^2/g$: $\phi'' = \dfrac{g}{r\omega^2}\sin\phi(\gamma\cos\phi - 1) = \sin\phi\Big(\cos\phi - \dfrac{1}{\gamma}\Big)$. (b) $\phi'' = -V'$ with $V = -\frac{1}{2}\sin^2\phi - \dfrac{\cos\phi}{\gamma}$. Equilibria: $\phi = 0, \pi$ always; $\cos\phi = 1/\gamma$ iff $\gamma > 1$. $F = \sin\phi(\cos\phi - 1/\gamma)$, $F' = \cos\phi(\cos\phi - 1/\gamma) - \sin^2\phi$.

- $\phi = \pi$: $F' = 1 + 1/\gamma > 0$: saddle for all $\gamma$.
- $\phi = 0$: $F' = 1 - 1/\gamma$: center for $\gamma < 1$, saddle for $\gamma > 1$.
- $\cos\phi_* = 1/\gamma$ ($\gamma > 1$): $F' = -\sin^2\phi_* < 0$: centers.

So two qualitatively different portraits, with a pitchfork at $\gamma = 1$. For $\gamma < 1$: a center at the bottom, a saddle at the top whose homoclinic orbit separates oscillations from full rotations. For $\gamma > 1$: saddle at the bottom with two homoclinic loops around the two new centers $\pm\phi_*$; beyond them larger orbits oscillate across the bottom, and then rotations beyond the separatrix through the top saddle. (c) Slow rotation ($\gamma < 1$): the bead oscillates about the bottom, or circulates if it has enough energy. Fast rotation ($\gamma > 1$): the bottom is unstable; the bead oscillates about an off-center equilibrium $\pm\phi_*$ on one side of the hoop, or with more energy across both, or circulates.




6.5.15: $\gamma < 1$ and $\gamma > 1$.

### 6.5.16 *Small oscillations*

**Solution.** With $b = 0$: $\ddot{\phi} = \sin\phi(\omega^2 \cos\phi - g/r) =: G(\phi)$. For $\omega^2 > g/r$ the symmetric stable equilibria are $\cos\phi_* = g/(r\omega^2)$. $G'(\phi_*) = -\omega^2 \sin^2\phi_*$, so small oscillations have

$$\Omega = \omega \sin\phi_* = \omega\sqrt{1 - \frac{g^2}{r^2\omega^4}} \quad \text{(in terms of } t\text{)}.$$

### 6.5.17 *A puzzling constant of motion*

**Solution.** Multiply $r\ddot{\phi} = -g\sin\phi + r\omega^2 \sin\phi\cos\phi$ by $\dot{\phi}$ and integrate:

$$J = \tfrac{1}{2}r\dot{\phi}^2 - g\cos\phi - \tfrac{1}{2}r\omega^2\sin^2\phi = \text{const} \quad \text{(constant of motion)},$$

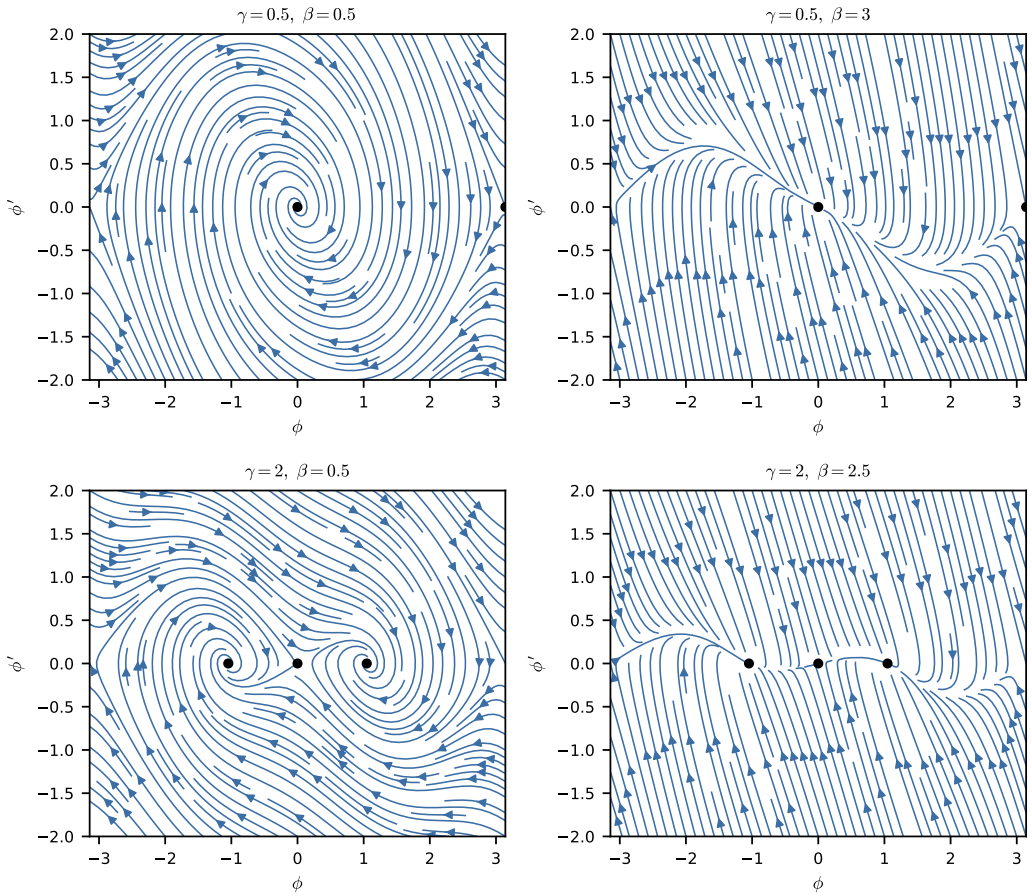
The bead's actual energy (per $m$) is $\mathcal{E} = \frac{1}{2}r^2\dot{\phi}^2 + \frac{1}{2}r^2\omega^2\sin^2\phi - gr\cos\phi$ (the second term is the kinetic energy of the circular motion about the rotation axis, $\frac{1}{2}(r\omega\sin\phi)^2$). where $\mathcal{E}$ is the bead's actual energy per unit mass. Then $\mathcal{E} - rJ = r^2\omega^2\sin^2\phi$, which is not constant, so $\mathcal{E}$ is not conserved. This makes sense: the hoop is driven at constant $\omega$ by an external agent, i.e. the constraint is time-dependent, and the constraint force does work on the bead. The conserved quantity $rJ$ is the energy in the *rotating frame*: kinetic energy relative to the hoop plus gravitational potential plus centrifugal potential $-\frac{1}{2}\omega^2(r\sin\phi)^2$ (the Jacobi integral).

### 6.5.18 *General damping*

**Solution.** Dimensionless form: $\phi'' = -\beta\phi' + \sin\phi\,(\cos\phi - 1/\gamma)$, with $\beta = \dfrac{b}{mr\omega}$ and $\gamma = \dfrac{r\omega^2}{g}$. Fixed points as in 6.5.15, but now $A = \begin{pmatrix} 0 & 1 \\ F' & -\beta \end{pmatrix}$, $\tau = -\beta$, $\Delta = -F'$. So saddles stay saddles ($\phi = \pi$ always; $\phi = 0$ for $\gamma > 1$) and centers become sinks: stable spirals if $\beta^2 < 4|F'|$, stable nodes if $\beta^2 > 4|F'|$:

- $\gamma < 1$: sink at $\phi = 0$ with $|F'| = 1/\gamma - 1$: spiral for $\beta < 2\sqrt{1/\gamma - 1}$, node otherwise.
- $\gamma > 1$: sinks at $\pm\phi_*$ with $|F'| = 1 - 1/\gamma^2$: spiral for $\beta < 2\sqrt{1 - 1/\gamma^2}$, node otherwise; saddle at $\phi = 0$.

So four portraits: ($\gamma < 1$ or $> 1$) $\times$ (underdamped spiral or overdamped node), with the pitchfork at $\gamma = 1$. The invariant manifolds of the saddle at the top spiral into the sinks when underdamped.



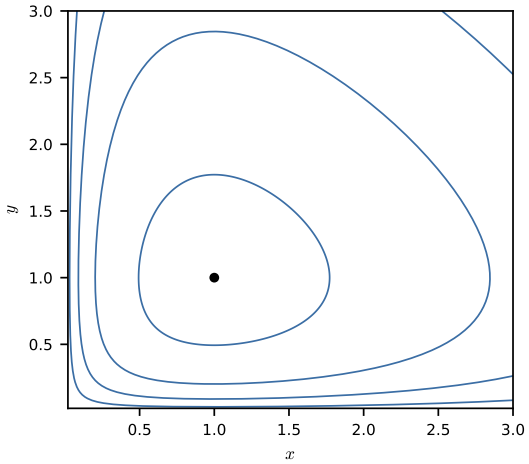

6.5.18: $(\gamma, \beta) = (0.5, 0.5), (0.5, 3), (2, 0.5), (2, 2.5)$.

### 6.5.19 *Lotka–Volterra:* $\dot{R} = aR - bRF,\ \dot{F} = -cF + dRF$

**Solution.** (a) $aR$: exponential rabbit growth without foxes (unlimited food, no carrying capacity); $-bRF$: predation, proportional to encounters; $-cF$: foxes starve at per-capita rate $c$ without rabbits; $+dRF$: foxes grow in proportion to rabbits eaten. Unrealistic: no prey saturation or self-limitation, no delays, no space or age structure. (b) $x = \dfrac{d}{c}R$, $y = \dfrac{b}{a}F$, $\tau = at$: $x' = x(1-y)$, $y' = \mu y(x-1)$, $\mu = c/a$. (c) $\dfrac{dy}{dx} = \dfrac{\mu y(x-1)}{x(1-y)} \Rightarrow \dfrac{1-y}{y}dy = \mu\dfrac{x-1}{x}dx$:

$$H(x, y) = \mu(x - \ln x) + y - \ln y = \text{const.}$$

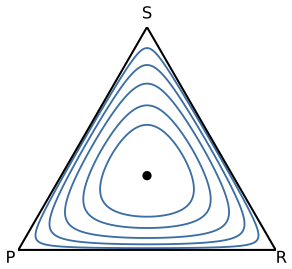
(Check: $\dot{H} = \mu\frac{x-1}{x}\,x(1-y) + \frac{y-1}{y}\,\mu y(x-1) = 0$.) (d) The fixed point $(1,1)$ (other: $(0,0)$ saddle). $H$ has a strict minimum at $(1,1)$ (Hessian $\operatorname{diag}(\mu, 1)$) and $H \to \infty$ as $x \to 0, \infty$ or $y \to 0, \infty$, so every level set is a compact closed curve containing no fixed point other than possibly $(1,1)$: every orbit in the open quadrant except $(1,1)$ is periodic. Predator and prey populations cycle forever.




6.5.19.

**6.5.20** *Rock–paper–scissors:* $\dot{P} = P(R - S),\ \dot{R} = R(S - P),\ \dot{S} = S(P - R)$

**Solution.** (a) $+PR$ in $\dot{P}$: paper gains when it meets rock (covers it); $-PS$: paper loses to scissors. $\dot{R}$: $+RS$ (rock smashes scissors), $-RP$ (loses to paper). $\dot{S}$: $+SP$ (scissors cut paper), $-SR$ (loses to rock). Implicit assumptions: random (mass-action) encounters in a well-mixed population; each encounter converts one individual with equal strength for all three pairs; no intrinsic growth, no carrying capacity, no within-species competition; continuous (large) populations. (b) $(P+R+S)^{\cdot} = P(R - S) + R(S - P) + S(P - R) = 0$. (c) $(PRS)^{\cdot} = PRS[(R - S) + (S - P) + (P - R)] = 0$. (d) Trajectories lie on the intersection of the plane $P + R + S = c_1$ and the surface $PRS = c_2$. On the plane, $PRS \le (c_1/3)^3$ with equality only at $P = R = S = c_1/3$ (AM–GM), and the level sets of $PRS$ in the triangle are closed curves around that point. Such a curve contains no fixed point (interior fixed points require $P = R = S$), so the trajectory, being confined to it and never stopping, runs around it periodically. Hence the populations cycle periodically forever; none goes extinct, since $PRS = c_2 > 0$ and $P + R + S = c_1$ keep each $\ge c_2/(c_1/2)^2 > 0$. The fixed point $P = R = S$ is a neutral center.



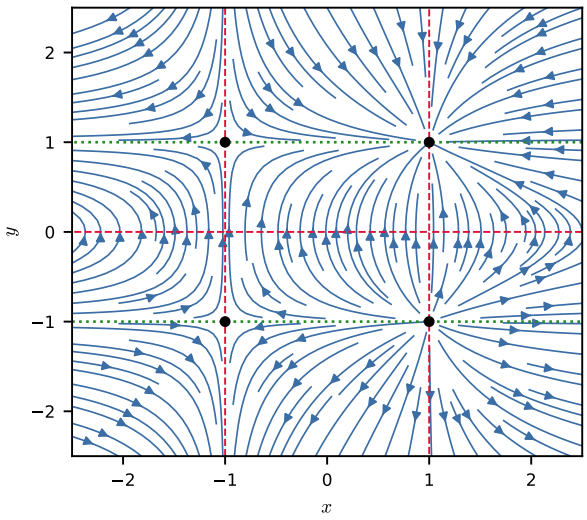

6.5.20: level curves of $PRS$ on the simplex $P + R + S = 1$.

## 6.6 Reversible Systems

Throughout, "reversible under $R$" means the equations are invariant under $t \to -t$ together with the stated change of variables; such systems cannot have attracting fixed points or limit cycles, and a linear center is a nonlinear center (Theorem 6.6.1).

### 6.6.1 $\dot{x} = y(1 - x^2),\ \dot{y} = 1 - y^2$

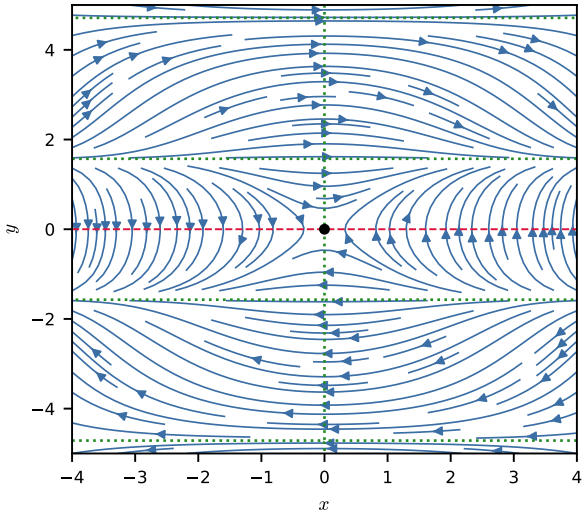
**Solution.** Under $t \to -t,\ y \to -y$: $\dot{x} \to -\dot{x} = -y(1 - x^2)$ ✓and $\dot{y} \to \dot{y} = 1 - y^2$ ✓: reversible. Fixed points $(\pm 1, \pm 1)$; $A = \begin{pmatrix} -2xy & 1 - x^2 \\ 0 & -2y \end{pmatrix}$: $(1, 1)$: $-2I$, stable star; $(1, -1)$: $+2I$, unstable star (the mirror images); $(-1, \pm 1)$: $\mathrm{diag}(\pm 2, \mp 2)$, saddles. The lines $x = \pm 1$ and $y = \pm 1$ are invariant; $y \to 1$ for $y > -1$ and $x \to 1$ if $x > -1$.




6.6.1.

### 6.6.2 $\dot{x} = y,\ \dot{y} = x\cos y$

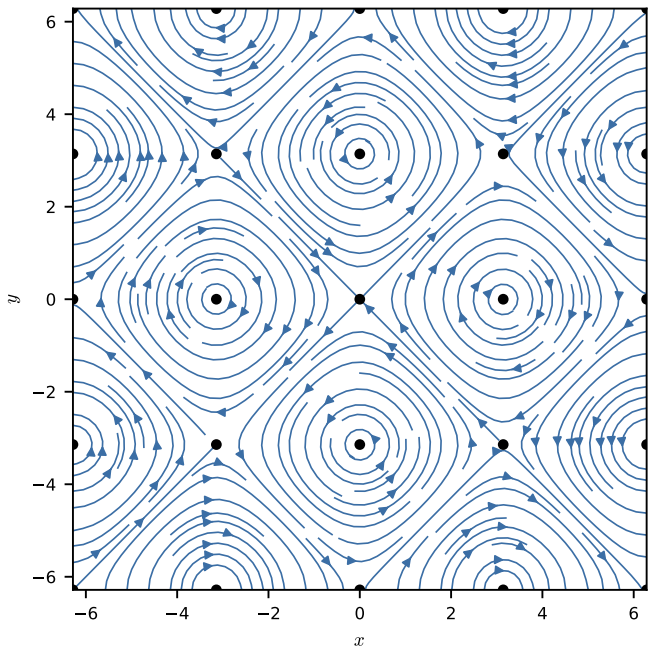
**Solution.** Reversible under $t \to -t, y \to -y$ (and also $t \to -t, x \to -x$). The only fixed point is the origin ($y = 0 \Rightarrow x = 0$), with $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: a saddle with eigen-directions $(1, \pm 1)$. The portrait is symmetric under reflection in both axes with arrows reversed.




6.6.2.

### 6.6.3 *Wallpaper:* $\dot{x} = \sin y,\ \dot{y} = \sin x$

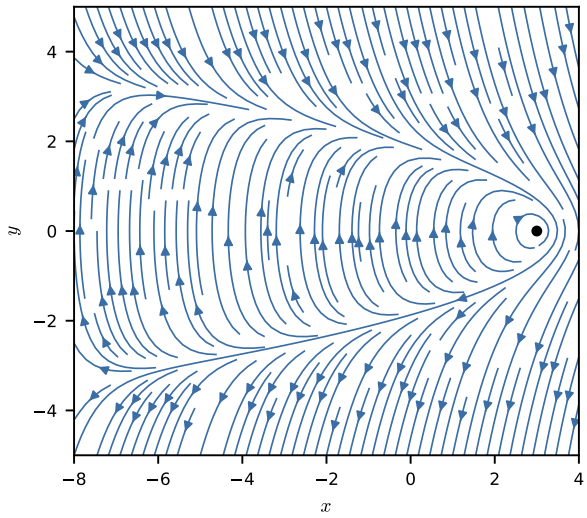
**Solution.** (a) Under $t \to -t, x \to -x$: $\dot{x} \to -\dot{x}$ vs. $\sin y$... precisely, $\frac{d(-x)}{d(-t)} = \dot{x} = \sin y$ ✓and $\frac{dy}{d(-t)} = -\dot{y} = -\sin x = \sin(-x)$ ✓. Same with $y \to -y$. (b) Fixed points $(m\pi, n\pi)$; $A = \begin{pmatrix} 0 & \cos y \\ \cos x & 0 \end{pmatrix}$, $\Delta = -(-1)^{m+n}$: saddles for $m + n$ even, centers for $m + n$ odd. $H = \cos x - \cos y$ is conserved, so the centers are nonlinear centers. (c) On $y = x$: $\dot{x} = \sin x = \dot{y}$, so $y - x$ stays 0; on $y = -x$: $\dot{x} = -\sin x = -\dot{y}$ so $x + y$ stays 0. (d) The diagonals $y = \pm x$ are heteroclinic separatrices between saddles; each diamond-shaped cell contains a center surrounded by closed orbits: a checkerboard pattern.



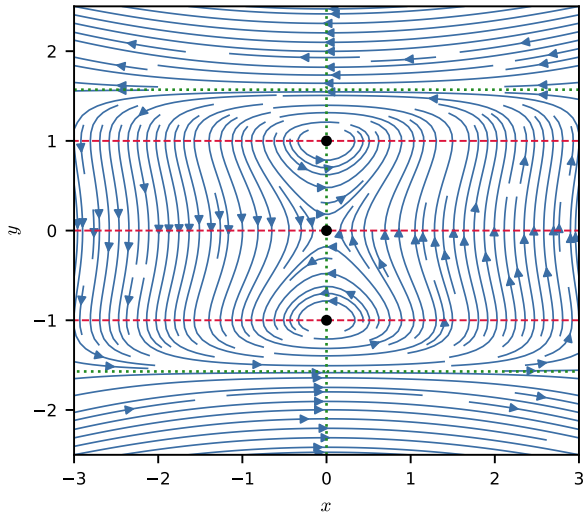

6.6.3.

### 6.6.4 *Computer explorations*

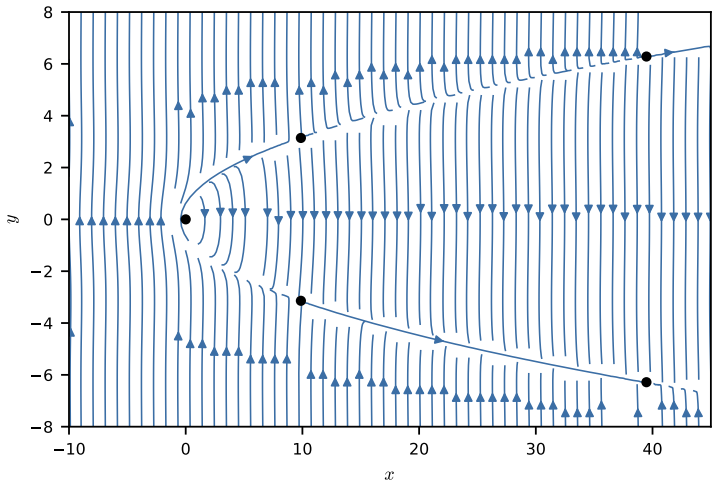
**Solution.** (a) $\ddot{x} + \dot{x}^2 + x = 3$: $\dot{x} = y$, $\dot{y} = 3 - x - y^2$, reversible under $y \to -y$. Fixed point $(3, 0)$, $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$: a nonlinear center. In fact $w = y^2$ satisfies $\frac{1}{2}dw/dx = 3 - x - w$, so $y^2 = Ce^{-2x} + \frac{7}{2} - x$. Closed orbits for $-\frac{1}{2}e^6 < C < 0$; they stretch leftward as $C \to 0^-$ and the family is bounded by the open parabola $y^2 = \frac{7}{2} - x$ ($C = 0$); orbits with $C > 0$ are open (they come from and return to $x = -\infty$). (b) $\dot{x} = y - y^3$, $\dot{y} = x \cos y$: reversible under $t \to -t, x \to -x$ (and $y \to -y$). Fixed points $(0, 0)$ ($\Delta = -1$, saddle) and $(0, \pm 1)$ ($\Delta = 2 \cos 1 > 0$: centers, nonlinear by reversibility). (c) $\dot{x} = \sin y$, $\dot{y} = y^2 - x$: reversible under $y \to -y$. Fixed points $(n^2\pi^2, n\pi)$, $A = \begin{pmatrix} 0 & (-1)^n \\ -1 & 2n\pi \end{pmatrix}$, $\tau = 2n\pi$, $\Delta = (-1)^n$: $n = 0$ center; $n$ odd saddles; $n = \pm 2, \pm 4, \ldots$ nodes (unstable for $n > 0$, stable for $n < 0$, mirror images of each other under the reversibility).




6.6.4(a).




6.6.4(b).




6.6.4(c).

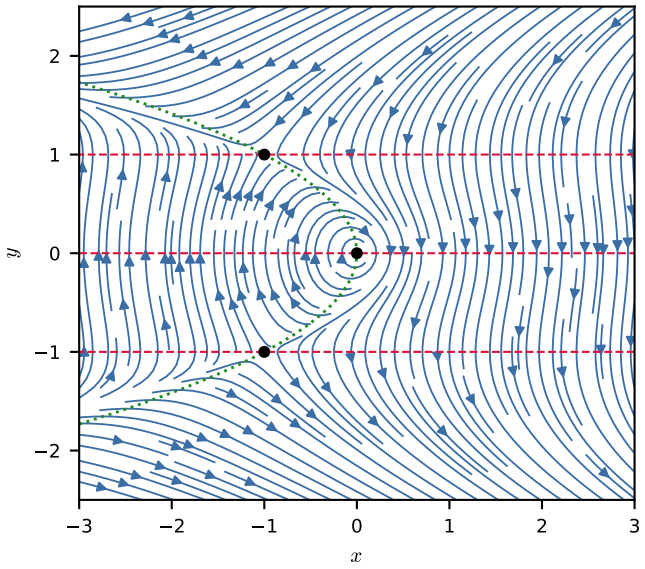
### 6.6.5 $\ddot{x} + f(\dot{x}) + g(x) = 0$, *f even*

**Solution.** (a) If $x(t)$ is a solution, put $X(t) = x(-t)$. Then $\dot{X} = \ddot{x}(-t)$, $\ddot{X} = -\dot{x}(-t)$, and $\ddot{X} + f(\dot{X}) + g(X) = \ddot{x}(-t) + f(-\dot{x}(-t)) + g(x(-t)) = \ddot{x}(-t) + f(\dot{x}(-t)) + g(x(-t)) = 0$ because $f$ is even. So the equation is invariant under $t \to -t$ (with $x$ unchanged, i.e. $y \to -y$ in the phase plane). (b) Equilibria: $y = 0, g(x^*) = 0$. $A = \begin{pmatrix} 0 & 1 \\ -g'(x^*) & -f'(0) \end{pmatrix}$ and $f'(0) = 0$ since $f$ is even and smooth. So $\tau = 0$, $\Delta = g'(x^*)$: eigenvalues $\pm\sqrt{-g'(x^*)}$, either real of opposite sign (saddle), or purely imaginary (center). Never a stable node or spiral.

### 6.6.6 *Manta ray:* $\dot{x} = y - y^3$, $\dot{y} = -x - y^2$

**Solution.** (a) Nullclines: $\dot{x} = 0$: $y = 0, \pm 1$ (horizontal lines); $\dot{y} = 0$: the parabola $x = -y^2$. (b) $\dot{x} > 0$ for $y < -1$ and $0 < y < 1$; $\dot{x} < 0$ for $-1 < y < 0$ and $y > 1$. $\dot{y} > 0$ to the left of the parabola ($x < -y^2$), $\dot{y} < 0$ to the right. (c) $A = \begin{pmatrix} 0 & 1 - 3y^2 \\ -1 & -2y \end{pmatrix}$. At $(-1, 1)$: $\begin{pmatrix} 0 & -2 \\ -1 & -2 \end{pmatrix}$, $\lambda = -1 \pm \sqrt{3}$: unstable $\lambda_1 = \sqrt{3} - 1$ with eigenvector $(1, -\frac{\sqrt{3}-1}{2}) \approx (1, -0.37)$, stable $\lambda_2 = -1 - \sqrt{3}$ with eigenvector $(1, 1.37)$. At $(-1, -1)$: $\begin{pmatrix} 0 & -2 \\ -1 & 2 \end{pmatrix}$, $\lambda = 1 \pm \sqrt{3}$: unstable $1 + \sqrt{3}$, eigenvector $(1, -1.37)$; stable $1 - \sqrt{3}$, eigenvector $(1, 0.37)$. (d) The unstable manifold of $(-1, -1)$ has a branch leaving in direction $(-1, 1.37)$ (up and to the left). In the strip $-1 < y < 0$: $\dot{x} < 0$ and, since $x < -1 < -y^2$, $\dot{y} > 0$. So $x$ decreases and $y$ increases (and $\dot{y} \geq -x - 1$ grows), hence $y$ reaches 0 in finite time: the branch crosses the negative $x$-axis at some $x_0 < -1$. By reversibility, reflecting across the $x$-axis and reversing time gives another trajectory through $(x_0, 0)$ which *arrives* at $(-1, 1)$ (the reflection of the unstable manifold of $(-1, -1)$ is the stable manifold of $(-1, 1)$). By uniqueness the two coincide: a heteroclinic trajectory from $(-1, -1)$ to $(-1, 1)$. (e) Consider the unstable manifold of $(-1, 1)$, branch in direction $(1, -0.37)$. It starts to the right of the parabola, in $0 < y < 1$ where $\dot{x} > 0$. The flow crosses the parabola $x = -y^2$ only from left to right there (on it $\dot{y} = 0, \dot{x} > 0$), so it stays right, where $\dot{y} < 0$. So $x$ increases, $y$ decreases and the branch reaches $y = 0$ at some $x_1 > 0$. By reversibility its mirror image is the stable manifold of $(-1, -1)$,

giving a second heteroclinic trajectory, now from $(-1, 1)$ to $(-1, -1)$ via the positive $x$-axis. The two heteroclinic orbits form the "manta-ray" outline, enclosing the nonlinear center at the origin, which is surrounded by closed orbits; outside the outline trajectories are unbounded.



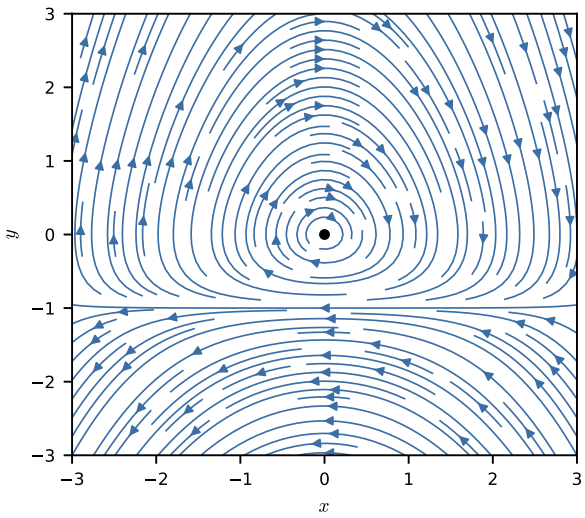

6.6.6: the manta ray.

### 6.6.7 $\ddot{x} + x\dot{x} + x = 0$

**Solution.** $\dot{x} = y,\ \dot{y} = -xy - x$. Under $t \to -t, x \to -x$: $\frac{d(-x)}{d(-t)} = \dot{x} = y$ ✓; $\frac{dy}{d(-t)} = -\dot{y} = xy + x$ and the right side becomes $-(-x)y - (-x) = xy + x$ ✓. Reversible. The origin is a linear center ($A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$), hence a nonlinear center. In fact $dy/dx = -x(1+y)/y$ gives $\dfrac{y}{1+y}dy = -x\,dx$, so

$$H = \tfrac{1}{2}x^2 + y - \ln|1 + y|$$

is conserved. The line $y = -1$ is invariant ($\dot{x} = -1, \dot{y} = 0$). For $y > -1$, $H \to \infty$ at the boundary $y \to -1^+$ and as $y \to \infty$, so every level set is closed: the whole half plane $y > -1$ is filled with closed orbits (positive damping for $x > 0$ cancels negative damping for $x < 0$). For $y < -1$ orbits are unbounded.

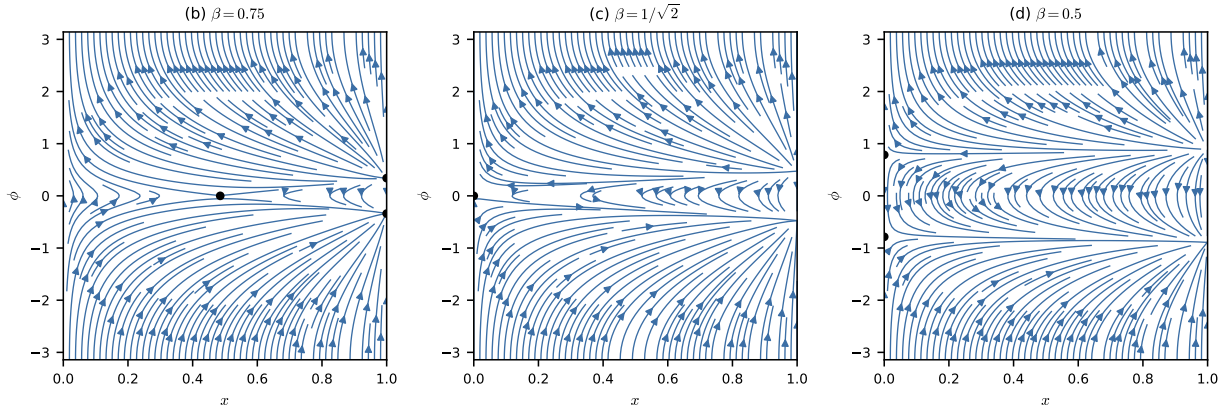



6.6.7.

### 6.6.8 *Reversible system on a cylinder*

**Solution.** $\dot{x} = \frac{\sqrt{2}}{4}x(x-1)\sin\phi$, $\dot{\phi} = \frac{1}{2}\big[\beta - \frac{1}{\sqrt{2}}\cos\phi - \frac{1}{8\sqrt{2}}x\cos\phi\big]$, $0 \le x \le 1$. (a) Under $t \to -t, \phi \to -\phi$: $\dot{x} \to -\dot{x}$ and $\sin\phi \to -\sin\phi$ ✓; $\dot{\phi}$ becomes $\frac{d(-\phi)}{d(-t)} = \dot{\phi}$ and the right side is even in $\phi$ ✓. Reversible. (b) Fixed points: $\dot{x} = 0$ iff $x = 0$, $x = 1$, or $\sin\phi = 0$. $\dot{\phi} = 0$ iff $\beta = \cos\phi\,\dfrac{8+x}{8\sqrt{2}}$. For $\frac{1}{\sqrt{2}} < \beta < \frac{9}{8\sqrt{2}}$: $x = 0$ would need $\cos\phi = \sqrt{2}\beta > 1$ (impossible); $\phi = \pi$ would need $\beta < 0$; $\phi = 0$ gives $x^* = 8\sqrt{2}\beta - 8 \in (0,1)$; $x = 1$ gives $\cos\phi_1 = 8\sqrt{2}\beta/9 < 1$, i.e. $\phi = \pm\phi_1$. Exactly three fixed points: $(x^*, 0)$, $(1, \pm\phi_1)$. *Saddle:* at $(x^*, 0)$, $A = \begin{pmatrix} 0 & \frac{\sqrt{2}}{4}x^*(x^*-1) \\ -\frac{1}{16\sqrt{2}} & 0 \end{pmatrix}$, both off-diagonal entries negative, so $\Delta < 0$: saddle. (At $(1, \pm\phi_1)$: $A$ is triangular with eigenvalues $\frac{\sqrt{2}}{4}\sin\phi$ and $\frac{9}{16\sqrt{2}}\sin\phi$ of the same sign: unstable node at $+\phi_1$, stable node at $-\phi_1$: mirror images.) *Homoclinic orbit and band of closed orbits.* For $x < x^*$: $\dot{\phi} \ge \frac{1}{2}\big[\beta - \frac{8+x}{8\sqrt{2}}\big] > 0$ for every $\phi$ (because $(8+x^*)/(8\sqrt{2}) = \beta$), so every trajectory with $x < x^*$ circles the cylinder monotonically. Start at $(x_0, 0)$ on the symmetry line $\phi = 0$ with $0 < x_0 < x^*$. It travels around and crosses the other symmetry line $\phi = \pi$ at some point. Reflecting the first half across $\phi = \pi$ (using reversibility $\phi \to 2\pi - \phi$, $t \to -t$) produces the second half, which returns to $(x_0, 0)$: the orbit is closed. This holds for every $x_0 \in (0, x^*)$, giving a band of closed orbits that wind around the waist of the cylinder, starting at the periodic orbit that is the circle $x = 0$ (on which $\dot{\phi} > 0$). The band is bounded by the unstable manifold of the saddle: eigenvalues $\pm\sqrt{AB}$ with eigenvector for the unstable one pointing toward decreasing $x$ and increasing $\phi$; this branch also circles the cylinder, crosses $\phi = \pi$, and by reversibility its mirror image is the saddle's stable manifold, so it comes back to the saddle: a homoclinic orbit around the waist. (c) As $\beta \to \frac{1}{\sqrt{2}}{}^+$, $x^* = 8\sqrt{2}\beta - 8 \to 0$: the saddle slides to the circle $x = 0$, squeezing the homoclinic loop (and the band) like a noose. At $\beta = 1/\sqrt{2}$ there are no fixed points inside $0 < x < 1$ ($x^* = 0$ lies on the boundary at $\phi = 0$), and $\dot{\phi} = \frac{1}{2\sqrt{2}}\big(1 - \cos\phi(1 + x/8)\big)$ is negative at $\phi = 0$ for $x > 0$ but positive at $\phi = \pi$, so no orbit can wind around the cylinder; a contractible closed orbit would have to enclose a fixed point (index theory). Hence all closed orbits are gone.

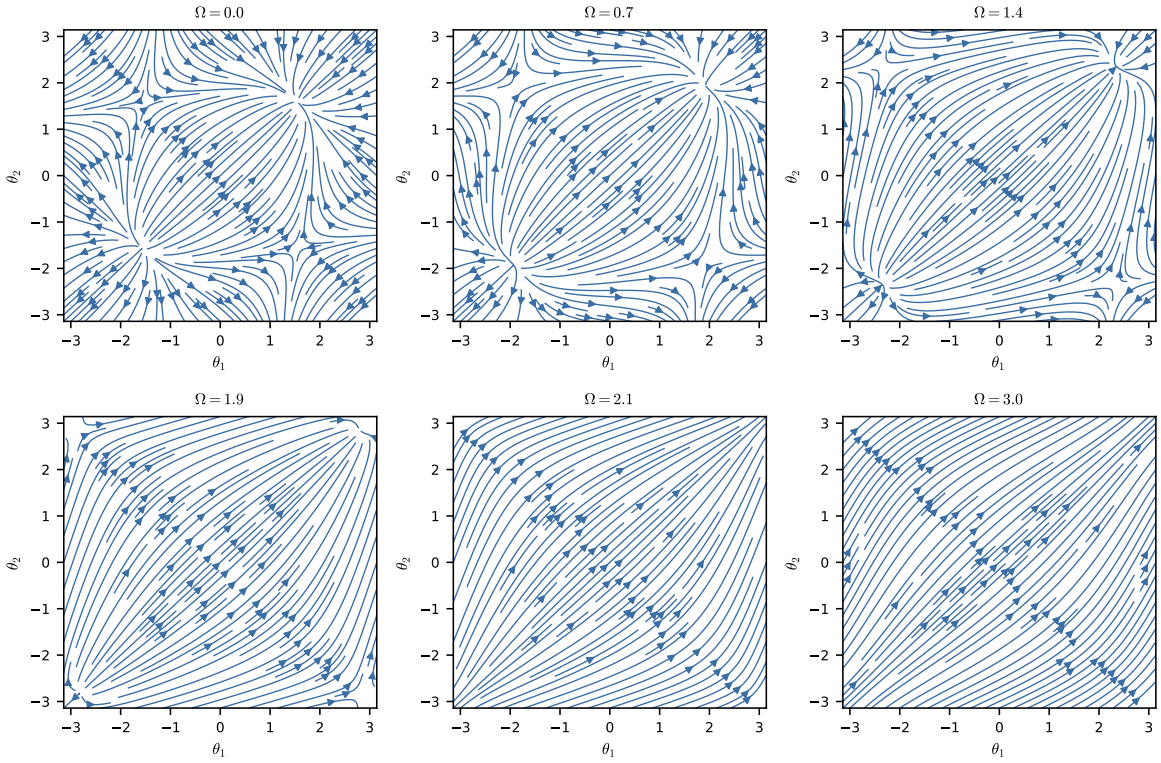
(d) For $0 < \beta < 1/\sqrt{2}$: $x = 0$ now gives $\cos\phi = \sqrt{2}\beta < 1$, two fixed points $(0, \pm\phi_0)$ on the edge, where $A$ is triangular with eigenvalues $\mp\frac{\sqrt{2}}{4}\sin\phi_0$ and $\pm\frac{1}{2\sqrt{2}}\sin\phi_0 = \pm\frac{\sqrt{2}}{4}\sin\phi_0$ of opposite signs: saddles. (And $\phi = 0$ would need $x^* < 0$: none.) The portrait is that shown in panel (d).




6.6.8: (b) $\beta = 0.75$, (c) $\beta = 1/\sqrt{2}$, (d) $\beta = 0.5$, drawn in $(x, \phi)$ coordinates with $\phi$ periodic.

### 6.6.9 *Josephson junction array ($N = 2$)*

**Solution.** $\dot\phi_k = \Omega + a\sin\phi_k + \frac{1}{2}(\sin\phi_1 + \sin\phi_2)$. (a) $\theta_k = \phi_k - \pi/2$: $\sin\phi_k = \cos\theta_k$, so $\dot\theta_k = \Omega + a\cos\theta_k + \frac{1}{2}(\cos\theta_1 + \cos\theta_2)$. Under $t \to -t$, $\theta_1 \to -\theta_2$, $\theta_2 \to -\theta_1$ the equations are invariant: e.g. $\frac{d(-\theta_2)}{d(-t)} = \dot\theta_2 = \Omega + a\cos\theta_2 + \frac{1}{2}(\cos\theta_1 + \cos\theta_2)$, which is the right side for $k = 1$ evaluated at the new variables (cosine is even). Reversible. (b) Subtracting the equations at a fixed point gives $a(\cos\theta_1 - \cos\theta_2) = 0$, so $\cos\theta_1 = \cos\theta_2 = c$, and then $\Omega + (a+1)c = 0$: $c = -\Omega/(a+1)$. Real solutions iff $|\Omega/(a+1)| \le 1$. For $|c| < 1$, $\theta_k = \pm\arccos c$ independently: four fixed points (mod $2\pi$). For $|\Omega/(a+1)| > 1$ none. (c) For $a = 1$ ($a + 1 = 2$): four fixed points for $\Omega < 2$ (for $\Omega = 0$ they are at $\theta_k = \pm\pi/2$: centers/saddles/nodes in the pattern forced by reversibility; as $\Omega \uparrow 2$ the fixed points merge in pairs and disappear). For $\Omega > 2$ there are no fixed points and all trajectories wind around the torus. The figure shows $\Omega = 0, 0.7, 1.4, 1.9$ (four fixed points), 2.1 and 3 (none).




6.6.9(c): $a = 1$.

### 6.6.10 $\dot{x} = -y - x^2,\ \dot{y} = x$

**Solution.** Linearization: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, a linear center. The system is reversible under $t \to -t, x \to -x$: $\frac{d(-x)}{d(-t)} = \dot{x} = -y - x^2$ (the right side is even in $x$) ✓; $\frac{dy}{d(-t)} = -\dot{y} = -x$ and the right side is $-x$ ✓. By Theorem 6.6.1, yes: the origin is a nonlinear center.

### 6.6.11 *Rotational dynamics and a phase portrait on a sphere*

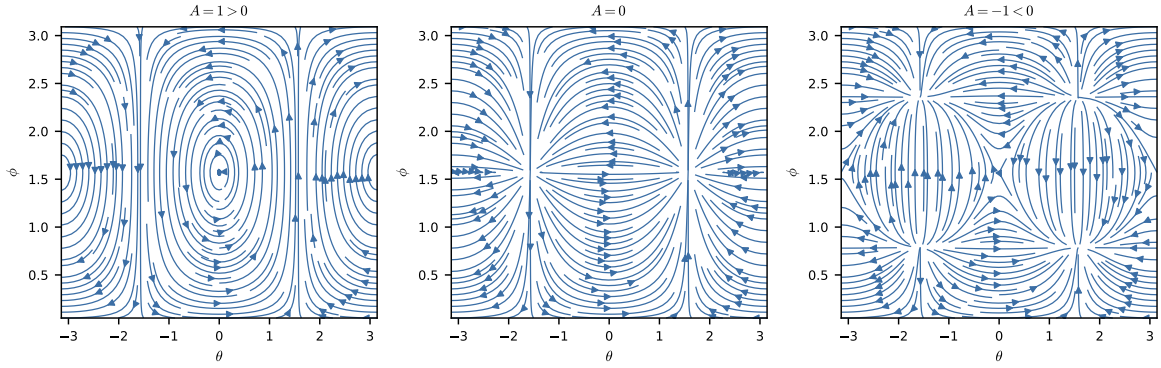
**Solution.** *Remark on the convention.* Taken literally with $\phi$ the latitude, $\dot{\theta} = \cot\phi\cos\theta$ blows up along the whole equator, which cannot describe a rigid rotation. The field is well behaved only if $\phi$ is measured from the pole (then $\cot\phi$ is singular only at the poles $\phi = 0, \pi$, a harmless coordinate singularity). This is also how the system arises from Jeffery's equations for a spheroid in simple shear (with $A$ the square of the aspect ratio). I therefore use $0 < \phi < \pi$ with $\phi = \pi/2$ the equator; the reflection symmetry $\phi \to -\phi$ of the text then reads $\phi \to \pi - \phi$ about the equator. (a) Under $t \to -t, \theta \to -\theta$: $\frac{d(-\theta)}{d(-t)} = \dot{\theta}$, the right side $\cot\phi\cos(-\theta)$ is unchanged ✓; $\frac{d\phi}{d(-t)} = -\dot{\phi}$ and $\sin(-\theta) = -\sin\theta$ ✓. Under $t \to -t, \phi \to \pi - \phi$ (reflection through the equator): cot changes sign so $\frac{d\theta}{d(-t)} = -\dot{\theta}$ ✓; $\frac{d(\pi-\phi)}{d(-t)} = \dot{\phi}$ and $\cos^2, \sin^2$ are unchanged ✓. (b) *Conserved quantity:* $\dfrac{d\phi}{d\theta} = \dfrac{(\cos^2\phi + A\sin^2\phi)\sin\theta\tan\phi}{\cos\theta}$ separates:

$$H = \frac{\sin\phi\cos\theta}{\sqrt{\cos^2\phi + A\sin^2\phi}} \quad \text{is constant, i.e.} \quad \tan^2\phi = \frac{h^2}{\cos^2\theta - Ah^2}.$$

*Fixed points* (on the equator $\phi = \pi/2$, $\theta = 0, \pi$): $A_{\text{Jac}} = \begin{pmatrix} 0 & \mp 1 \\ \pm A & 0 \end{pmatrix}$ with $\Delta = A$. The poles are regular points where the flow has a nonzero velocity (they are not fixed points).

- $A > 0$: centers at $(0, \pi/2)$ and $(\pi, \pi/2)$; all orbits are closed: loops around these two points $(0 < |h| < 1/\sqrt{A})$, and the great circle $\theta = \pm\pi/2$ through both poles ($h = 0$), traversed pole to pole.
- $A = 0$: $\tan\phi \cos\theta = h$ and the whole equator $\phi = \pi/2$ is a line of fixed points; every other orbit is an arc running from an equator point $\theta = -\pi/2$ to an equator point $\theta = \pi/2$; the great circle through the poles is an orbit.
- $A < 0$: saddles at $(0, \pi/2), (\pi, \pi/2)$ and four more fixed points $(\pm\pi/2, \phi_A)$, $(\pm\pi/2, \pi - \phi_A)$, $\tan^2\phi_A = 1/|A|$: diagonal Jacobian with eigenvalues $-\cot\phi \sin\theta$ and $-(1 - A)\sin 2\phi \sin\theta$ of equal sign, so nodes: stable at $(\pi/2, \phi_A)$ and $(-\pi/2, \pi - \phi_A)$, unstable at the other two.

(c) $A > 0$ (the case for real spheroids): every orientation (except the two centers) is periodic: the object *tumbles* forever, never settling. $A = 0$: the orientation tends to a point on the equator ($\phi = \pi/2$) and the object stops tumbling. $A < 0$: the orientation tends to one of the two stable nodes: a steady alignment, and tumbling ceases.

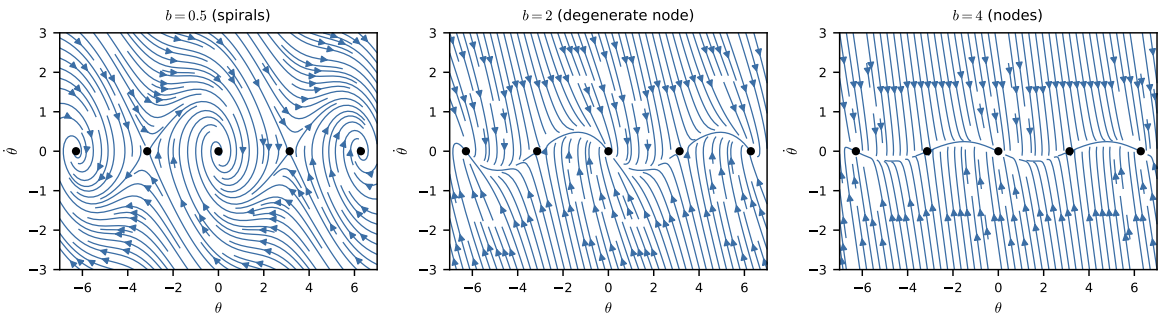



6.6.11 (Mercator view, $\theta \in [-\pi, \pi]$, $\phi \in (0, \pi)$): $A = 1, 0, -1$.

## 6.7 Pendulum

### 6.7.1 *Damped pendulum* $\ddot{\theta} + b\dot{\theta} + \sin\theta = 0$, $b > 0$

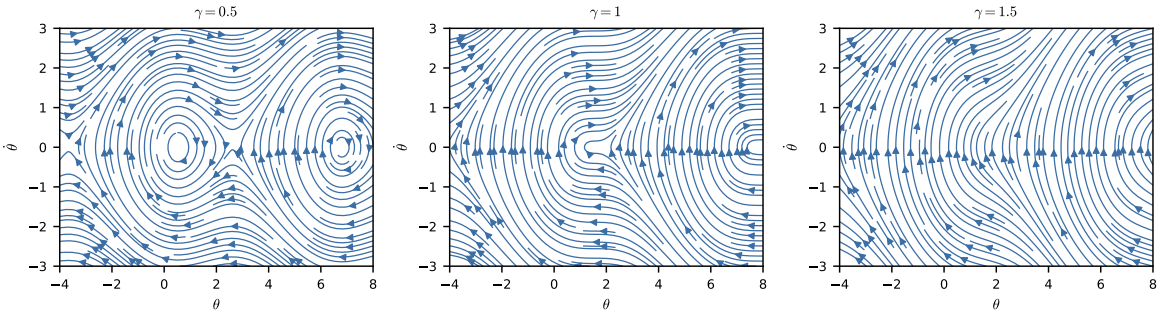
**Solution.** $\dot{\theta} = v$, $\dot{v} = -bv - \sin\theta$. Fixed points $(n\pi, 0)$, $A = \begin{pmatrix} 0 & 1 \\ -\cos\theta & -b \end{pmatrix}$. $\theta = 2k\pi$: $\tau = -b, \Delta = 1$: always stable; $\lambda = \frac{-b \pm \sqrt{b^2 - 4}}{2}$: stable spiral for $b < 2$, stable degenerate node for $b = 2$, stable node for $b > 2$. $\theta = (2k + 1)\pi$: $\Delta = -1$: saddles. Three qualitatively different portraits: underdamped ($b < 2$), critically damped ($b = 2$), overdamped ($b > 2$).




6.7.1.

### 6.7.2 *Pendulum driven by constant torque:* $\ddot\theta + \sin\theta = \gamma$

**Solution.** (a) $\dot\theta = v, \dot v = \gamma - \sin\theta$. Equilibria: $\sin\theta = \gamma$. For $|\gamma| > 1$ none. For $|\gamma| < 1$: $\theta_1 = \arcsin\gamma$ and $\theta_2 = \pi - \arcsin\gamma \pmod{2\pi}$. $A = \begin{pmatrix} 0 & 1 \\ -\cos\theta & 0 \end{pmatrix}$, $\Delta = \cos\theta$: $\theta_1$: $\Delta = \sqrt{1-\gamma^2} > 0$, center; $\theta_2$: saddle. At $|\gamma| = 1$ they merge: a saddle-node (degenerate). (b) Nullclines: $v = 0$ and the vertical lines $\sin\theta = \gamma$. For $v > 0$ the flow goes right; $\dot v > 0$ where $\sin\theta < \gamma$. (c) Conservative: $E = \frac{1}{2}v^2 - \cos\theta - \gamma\theta$ ($\ddot\theta = -V'$, $V = -\cos\theta - \gamma\theta$; on the cylinder $E$ is multivalued). Reversible under $t \to -t, v \to -v$. (d) $|\gamma| < 1$: a row of centers each surrounded by closed orbits and bounded by homoclinic loops through the saddles; above and below, unbounded orbits (the pendulum rotating). $\gamma = 1$: the centers and saddles collide. $|\gamma| > 1$: no equilibria; $\dot v \geq \gamma - 1 > 0$, so every trajectory moves upward, parabola-like curves $\theta \approx v^2/(2\gamma)$. (e) Small oscillations about $\theta_1$: $\omega^2 = \cos\theta_1$, so $\omega = (1-\gamma^2)^{1/4}$.



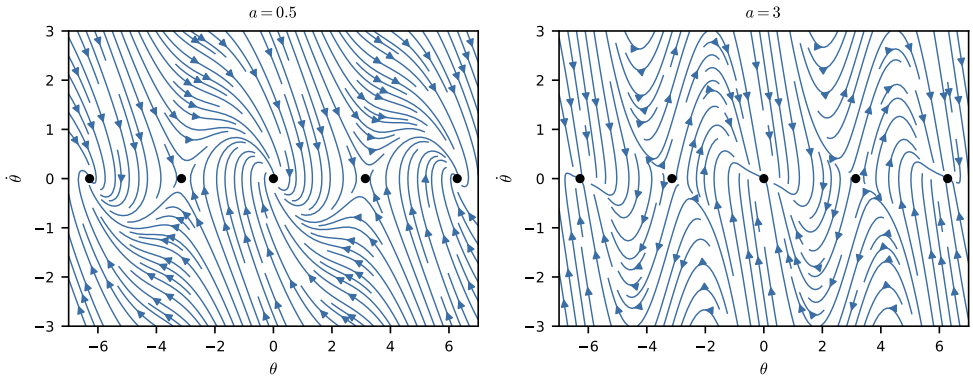

6.7.2: $\gamma = 0.5, 1, 1.5$.

### 6.7.3 *Nonlinear damping:* $\ddot\theta + (1 + a\cos\theta)\dot\theta + \sin\theta = 0$, $a \geq 0$

**Solution.** $\dot\theta = v$, $\dot v = -(1 + a\cos\theta)v - \sin\theta$. Fixed points $(n\pi, 0)$, $A = \begin{pmatrix} 0 & 1 \\ -\cos\theta & -(1 + a\cos\theta) \end{pmatrix}$.

- $\theta = 2k\pi$: $\tau = -(1+a) < 0$, $\Delta = 1$: stable; spiral for $a < 1$, degenerate node for $a = 1$, node for $a > 1$.
- $\theta = (2k+1)\pi$: $\Delta = -1$: saddle for all $a$.

For $a > 1$ the damping coefficient $1 + a\cos\theta$ becomes negative near the top ($\cos\theta < -1/a$): the pendulum is pumped there. Numerical integration (e.g. $a = 1.5, 3, 6, 12$, started in rotation) shows that the trajectories nevertheless all settle on the stable fixed points; I found no rotating limit cycle (I have not proved their absence).




6.7.3: $a = 0.5$ and $a = 3$.

### 6.7.4 *Period of the pendulum*

**Solution.** (a) $E = \frac{1}{2}\dot\theta^2 - \cos\theta$ is conserved. At the amplitude $\alpha$, $\dot\theta = 0$, so $E = -\cos\alpha$ and $\dot\theta^2 = 2(\cos\theta - \cos\alpha)$. The period is four times the time to go from 0 to $\alpha$: $T = 4\int_0^\alpha d\theta/[2(\cos\theta - \cos\alpha)]^{1/2}$. (b) $\cos\theta = 1 - 2\sin^2\frac{\theta}{2}$: $2(\cos\theta - \cos\alpha) = 4(\sin^2\frac{\alpha}{2} - \sin^2\frac{\theta}{2})$, so $T = 4\int_0^\alpha \dfrac{d\theta}{[4(\sin^2\frac{\alpha}{2} - \sin^2\frac{\theta}{2})]^{1/2}}$. (c) Put $k = \sin\frac{\alpha}{2}$ and $k\sin\varphi = \sin\frac{\theta}{2}$ ($\varphi : 0 \to \pi/2$ as $\theta : 0 \to \alpha$). Then $k\cos\varphi\, d\varphi = \frac{1}{2}\cos\frac{\theta}{2}\, d\theta$ and the denominator is $2k\cos\varphi$, so

$$T = 4\int_0^{\pi/2} \frac{d\varphi}{\cos\frac{\theta}{2}} = 4\int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2\sin^2\varphi}} = 4K\left(\sin^2\tfrac{\alpha}{2}\right).$$

(d) $K(m) = \int_0^{\pi/2}\left[1 + \frac{1}{2}m\sin^2\varphi + \frac{3}{8}m^2\sin^4\varphi + \dots\right]d\varphi = \frac{\pi}{2}\left[1 + \frac{m}{4} + \frac{9m^2}{64} + \dots\right]$. With $m = \sin^2\frac{\alpha}{2} = \frac{\alpha^2}{4} + O(\alpha^4)$:

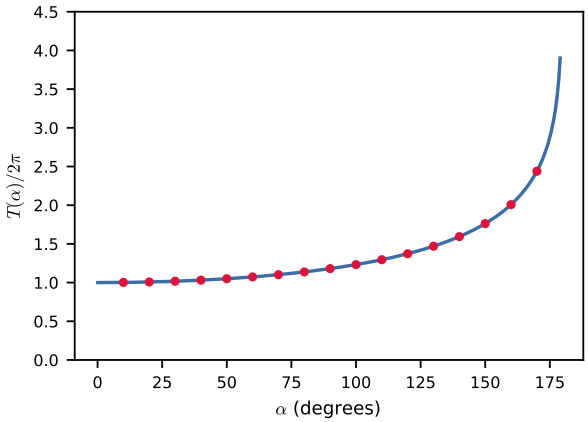
$$T(\alpha) = 2\pi\Big[1 + \tfrac{1}{16}\alpha^2 + O(\alpha^4)\Big],$$

so larger swings take longer.

### 6.7.5 *Numerical period*

**Solution.** $T(\alpha) = 4K(\sin^2\frac{\alpha}{2})$ evaluated with the complete elliptic integral (and cross-checked by integrating $\ddot\theta = -\sin\theta$ numerically and timing the return, which agreed to all digits shown). $T(180^\circ) = \infty$: starting at the unstable equilibrium, the pendulum never leaves.

| $\alpha$ (deg) | $T(\alpha)$ | $T/2\pi$ |
|---|---|---|
| 0 | 6.2832 | 1.0000 |
| 10 | 6.2952 | 1.0019 |
| 20 | 6.3314 | 1.0077 |
| 30 | 6.3926 | 1.0174 |
| 40 | 6.4801 | 1.0313 |
| 50 | 6.5960 | 1.0498 |
| 60 | 6.7430 | 1.0732 |
| 70 | 6.9250 | 1.1021 |
| 80 | 7.1471 | 1.1375 |
| 90 | 7.4163 | 1.1803 |
| 100 | 7.7423 | 1.2322 |
| 110 | 8.1389 | 1.2953 |
| 120 | 8.6261 | 1.3729 |
| 130 | 9.2351 | 1.4698 |
| 140 | 10.0182 | 1.5944 |
| 150 | 11.0723 | 1.7622 |
| 160 | 12.6135 | 2.0075 |
| 170 | 15.3270 | 2.4394 |
| 180 | $\infty$ | $\infty$ |




6.7.5: $T(\alpha)/2\pi$.

## 6.8 Index Theory

### 6.8.1 *Fixed points with index* 1

**Solution.** Near a linear fixed point $\dot{\mathbf{x}} = A\mathbf{x}$, the index is the winding number of the vector $A\mathbf{x}$ as $\mathbf{x}$ goes once counterclockwise around a circle. $A$ maps the circle to an ellipse around the origin; the orientation is preserved iff $\det A > 0$, in which case $A\mathbf{x}$ winds $+1$ time. Stable and unstable spirals, centers, stars and degenerate nodes all have $\Delta = \lambda_1\lambda_2 > 0$ (complex pair, or real eigenvalues of

the same sign), so each has index +1. (Saddles have $\Delta < 0$, index −1.) Direct check: the center $\dot{x} = -y, \dot{y} = x$ and the star $\dot{x} = x, \dot{y} = y$ both have field direction $\theta + \pi/2$ or $\theta$, rotating once.

### 6.8.2 $\dot{x} = x^2,\ \dot{y} = y$

**Solution.** Fixed point: origin. On the circle, the field $(\cos^2\theta, \sin\theta)$ has non-negative $x$-component, so its direction stays in $[-\pi/2, \pi/2]$ and never winds: **index** 0.

### 6.8.3 $\dot{x} = y - x,\ \dot{y} = x^2$

**Solution.** Origin only. On the circle the field is $(\sin\theta - \cos\theta, \cos^2\theta)$, whose $y$-component is $\geq 0$: the direction stays in $[0, \pi]$: **index** 0.

### 6.8.4 $\dot{x} = y^3,\ \dot{y} = x$

**Solution.** Origin only. Field $(\sin^3\theta, \cos\theta)$: direction $\pi/2$ at $\theta = 0$, 0 at $\pi/2$, $-\pi/2$ at $\pi$, $-\pi$ at $3\pi/2$, $-3\pi/2$ at $2\pi$: it decreases monotonically by $2\pi$: **index** −1 (a saddle-like point).

### 6.8.5 $\dot{x} = xy,\ \dot{y} = x + y$

**Solution.** Origin only ($xy = 0$ and $x + y = 0$). Field on the circle: $(\cos\theta \sin\theta, \cos\theta + \sin\theta)$. Its direction is 90° at $\theta = 0$, about 70° at 45°, 90° at 90°, 180° at 135°, 270° at 180°, about 290° at 225°, 270° at 270°, 180° at 315°, 90° at 360°: it goes up and comes back down with no net winding: **index** 0.

### 6.8.6 *A closed orbit enclosing $S$ saddles, $N$ nodes, $F$ spirals, $C$ centers*

**Solution.** The index of a closed orbit is +1 and equals the sum of the indices inside: $+1 = -S + N + F + C$, hence $N + F + C = 1 + S$.

### 6.8.7 $\dot{x} = x(4 - y - x^2),\ \dot{y} = y(x - 1)$

**Solution.** Fixed points: $(0, 0)$ (saddle, eigenvalues $4, -1$), $(\pm 2, 0)$ ($(2, 0)$: $\begin{pmatrix} -8 & -2 \\ 0 & 1 \end{pmatrix}$ saddle; $(-2, 0)$: eigenvalues $-8, -3$ stable node), and $(1, 3)$ ($\begin{pmatrix} -2 & -1 \\ 3 & 0 \end{pmatrix}$, stable spiral). Both coordinate axes are invariant, so a closed orbit cannot cross them and lies in one open quadrant; it must enclose fixed points of total index +1. The second, third and fourth quadrants contain no fixed point in their interiors (the points on the axes cannot be enclosed by a curve inside the open quadrant), which rules out closed orbits there. In the open first quadrant the only fixed point is $(1, 3)$, with index +1, so index theory alone permits a closed orbit around it. To finish: use Dulac's criterion (Section 7.2) with $g = 1/(xy)$ in $x, y > 0$: $\nabla \cdot (g\dot{\mathbf{x}}) = \partial_x \frac{4-y-x^2}{y} + \partial_y \frac{x-1}{x} = -\frac{2x}{y} < 0$, so there is no closed orbit in the first quadrant either. (Index theory narrows the possibilities; Dulac's criterion removes the last one.)

### 6.8.8 *Three closed orbits; $C_1, C_2$ inside $C_3$, neither inside the other*

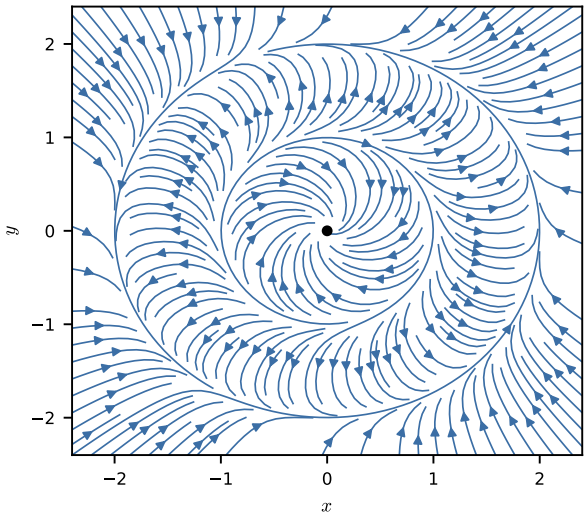
**Solution.** (a) A large loop $C_3$ containing two disjoint smaller loops $C_1$ and $C_2$ side by side. (b) Each closed orbit has index +1. The index of $C_3$ equals the sum of the indices of all fixed points inside it: $+1 = I(C_1) + I(C_2) + I_{\text{between}} = 1 + 1 + I_{\text{between}}$. So the fixed points in the region between $C_1, C_2, C_3$ have total index $-1 \neq 0$: there must be at least one (e.g. a saddle).

### 6.8.9 *Two nested closed orbits, inner clockwise, outer counterclockwise*

**Solution. False.** The index of a closed orbit is $+1$ regardless of the direction of travel (reversing the field rotates every vector by $\pi$ and does not change the winding number), so the indices of the inner and outer orbit agree and the region between need not contain a fixed point. Counterexample (polynomial, smooth):

$$\dot r = r(r^2-1)(4-r^2), \qquad \dot\theta = r^2 - \tfrac{5}{2},$$

i.e. $\dot x = x(r^2-1)(4-r^2) - y(r^2-\frac{5}{2})$, $\dot y = y(r^2-1)(4-r^2) + x(r^2-\frac{5}{2})$. Closed orbits: $r=1$ ($\dot\theta = -\frac{3}{2} < 0$, clockwise) and $r=2$ ($\dot\theta = \frac{3}{2} > 0$, counterclockwise). Between them $\dot r > 0$, so there is no fixed point there; $\dot\theta$ vanishes at $r^2 = 5/2$ but $\dot r \neq 0$ there. The only fixed point is the origin, inside the inner cycle.




6.8.9.

### 6.8.10 *Theorem 6.8.2 on other surfaces*

**Solution.** The statement "index of a closed orbit $= +1 =$ sum of indices of fixed points inside" needs the orbit to bound a disk. *Cylinder:* a closed orbit that winds around the cylinder bounds no disk, and the theorem fails: $\dot\theta = 1, \dot y = 0$ has circles as closed orbits and no fixed points. A contractible closed orbit (bounding a disk) does satisfy the theorem. *Torus:* likewise, orbits winding around either hole may have no fixed points ($\dot\theta_1 = \dot\theta_2 = 1$ has closed orbits if the frequencies are commensurate). A contractible closed orbit bounds a disk and the theorem holds for that disk; the fixed points outside have total index $-1$ (the total index on the torus is $\chi = 0$). *Sphere:* a closed orbit divides the sphere into two disks, and the theorem applies to both: each side contains fixed points of total index $+1$, total $2 = \chi(S^2)$ (Poincaré–Hopf).

### 6.8.11 *Complex vector fields $\dot z = z^k$ and $\dot z = (\bar z)^k$*

**Solution.** (a) Polar: $\dot z = (\dot r + ir\dot\theta)e^{i\theta}$. For $z^k = r^k e^{ik\theta}$: $\dot r = r^k\cos((k-1)\theta)$, $\dot\theta = r^{k-1}\sin((k-1)\theta)$. For $\bar z^k = r^k e^{-ik\theta}$: $\dot r = r^k\cos((k+1)\theta)$, $\dot\theta = -r^{k-1}\sin((k+1)\theta)$. Cartesian, $\dot z = z^k$: $k=1$: $\dot x = x, \dot y = y$; $k=2$: $\dot x = x^2 - y^2, \dot y = 2xy$; $k=3$: $\dot x = x^3 - 3xy^2, \dot y = 3x^2y - y^3$. Cartesian, $\dot z = \bar z^k$:

$k=1$: $\dot x = x, \dot y = -y$; $k=2$: $\dot x = x^2 - y^2, \dot y = -2xy$; $k=3$: $\dot x = x^3 - 3xy^2, \dot y = y^3 - 3x^2y$. (b) $z^k = 0$ only at $z = 0$, so the origin is the only fixed point. On the unit circle the field is $e^{ik\theta}$, whose argument is $k\theta$: it winds $k$ times, so the index is $k$ for $\dot z = z^k$ and $-k$ for $\dot z = \bar z^k$ (argument $-k\theta$). (c) For every integer $k > 0$: index $+k$ for $z^k$, index $-k$ for $\bar z^k$. ($k = 1$: star, index $+1$; saddle, index $-1$. $k = 2$ is the dipole-type point of 6.1.9, index 2.)

### 6.8.12 *"Matter and antimatter": $\dot x = a + x^2,\ \dot y = -y$*

**Solution.** (a) For $a < 0$: fixed points $(\pm\sqrt{-a}, 0)$; $A = \operatorname{diag}(2x, -1)$: $(\sqrt{-a}, 0)$ is a saddle, $(-\sqrt{-a}, 0)$ a stable node. At $a = 0$ they merge into one semi-stable (saddle-node) point $(0, 0)$ with $A = \operatorname{diag}(0, -1)$. For $a > 0$ there are none. (b) Indices: $-1 + 1 = 0$ for $a < 0$; 0 at $a = 0$ (as in 6.8.2); 0 (no fixed points) for $a > 0$. The total index is conserved. (c) *Claim:* for $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, a)$, $\mathbf{x} \in \mathbb{R}^2$, $\mathbf{f}$ continuous in $a$, let $C$ be a simple closed curve containing no fixed points for $a \in [a_1, a_2]$. Then the sum of the indices of the fixed points inside $C$ is the same for all $a \in [a_1, a_2]$. *Proof:* the index $I_C(a)$ is the winding number of $\mathbf{f}(\cdot, a)$ along $C$; it is an integer, depends continuously on $a$ (since $\mathbf{f} \neq 0$ on $C$), hence constant; and it equals the sum of indices of the fixed points inside. So fixed points can only be created or annihilated inside $C$ in groups of total index zero (e.g. a saddle and a node), or enter/leave through $C$.

### 6.8.13 *Integral formula for the index*

**Solution.** (a) With $\phi = \tan^{-1}(g/f)$, $d\phi = \dfrac{1}{1 + g^2/f^2}\, d\!\left(\dfrac{g}{f}\right) = \dfrac{f^2}{f^2+g^2} \cdot \dfrac{f\,dg - g\,df}{f^2} = \dfrac{f\,dg - g\,df}{f^2+g^2}$.
(b) $I_C = \dfrac{1}{2\pi} \oint_C d\phi = \dfrac{1}{2\pi} \oint_C \dfrac{f\,dg - g\,df}{f^2+g^2}$, where $df = f_x dx + f_y dy$, $dg = g_x dx + g_y dy$ along $C$.

### 6.8.14 *Rotation family $\dot x = x\cos\alpha - y\sin\alpha,\ \dot y = x\sin\alpha + y\cos\alpha$*

**Solution.** (a) $A = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$, $\lambda = e^{\pm i\alpha}$, $\tau = 2\cos\alpha$, $\Delta = 1$. $\alpha = 0$: $A = I$, unstable star; $0 < \alpha < \pi/2$: unstable spiral; $\alpha = \pi/2$: center; $\pi/2 < \alpha < \pi$: stable spiral; $\alpha = \pi$: $A = -I$, stable star. (b) For all $\alpha$ the only fixed point is the origin ($\Delta = 1$), so on a curve $C$ avoiding the origin, $f, g$ do not vanish together, the integrand in 6.8.13 is continuous in $\alpha$, and $I_C(\alpha)$, an integer, is independent of $\alpha$. (c) Take $\alpha = \pi/2$ ($f = -y, g = x$) and $C$ a circle of radius $\rho$: $f\,dg - g\,df = -y\,dx + x\,dy = \rho^2 dt$ and $f^2 + g^2 = \rho^2$, so $I_C = \frac{1}{2\pi}\int_0^{2\pi} dt = 1$.